

Review

Machine learning and hybrid intelligence for wind energy optimization: A comprehensive state-of-the-art review

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ABSTRACT

Wind energy plays a pivotal role in the global transition toward sustainable energy. However, its intermittent and stochastic nature presents challenges in achieving optimal performance, reliability, and seamless grid integration. Recent advances in machine intelligence—including machine learning (ML), deep learning (DL), and reinforcement learning (RL)—offer powerful tools to address these challenges across forecasting, control,

Abbreviations: 2-DOF, Two-degree-of-freedom; 5G-NG-RAN, 5G-Next Generation-Radio Access Network; AI, Artificial Intelligence; AD-PSO-WOA, Adaptive Differential PSO-Guided Whale Optimization Algorithm; AEP, Annual Energy Production; AGA, Adaptive Genetic Algorithm; AGA-DDPG, Adaptive Genetic Algorithm + Deep Deterministic Policy Gradient; ANFIS, Adaptive Neuro-Fuzzy Inference System; AUC, Area Under the Curve; BPNN, Back-Propagation Neural Network; CFD, Computational Fluid Dynamics; CI/CD, Continuous Integration and Continuous Deployment; CNN, Convolutional Neural Network; CVSS, Common Vulnerability Scoring System; DBN, Deep Belief Network; DCNN, Deep CNN; DFIG, Doubly Fed Induction Generators; D-GABC, Density-based Grey Artificial Bee Colony; DQN, Deep Q-network; DDQN, Double DQN; DL, Deep Learning; DNN, Deep Neural Network; DRL, Deep Reinforcement Learning; DT, Decision Tree; EEMD, Ensemble Empirical Mode Decomposition; EMD, Empirical Mode Decomposition; ELM, Extreme Learning Machine; EMS, Energy Management System; ESS, Energy Storage System; EWT, Empirical Wavelet Transform; FFT, Fast Fourier Transform; GAN, Generative Adversarial Networks; GGWO, Genetic Grey Wolf Optimizer; GP, Gaussian Processes; GMDH, Group Method of Data Handling; GRNN, Generalized Regression Neural Network; GRU, Gated Recurrent Unit; GWO, Grey Wolf Optimization; HARO, Huber Adam Regression Optimizer; HHO, Harris Hawks Optimization; IGBT, Insulated Gate Bipolar Transistor; IoT, Internet of Things; JVAE, Joint Variational Autoencoder; KNN, K-Nearest Neighbor; LASSO, Least Absolute Shrinkage and Selection Operator; LC, Lagrangian Control; LiDAR, Light Detection and Ranging; LDA, Linear Discriminant Analysis; LR, Linear Regression; LIME, Local Interpretable Model-agnostic Explanations; LSTM, Long Short-Term Memory; LUBE, Lower Upper Bound Estimation; MAE, Mean Absolute Error; MAPE, Mean Absolute Percentage Error; MCP, Measure-Correlate-Predict; MEC, Multi-access Edge Computing; Meta-MARL, Meta Multi-Agent Reinforcement Learning; ML, Machine Learning; MLP, Multi-Layer Perceptron; MPC, Model Predictive Control; MPPT, Maximum Power Point Tracking; MQ, Mapping Questions; NARX, Nonlinear Autoregressive Exogenous; NB, Naive Bayes; NN, Neural Network; NMAE, Normalized Mean Absolute Error; NRMSE, Normalized Root Mean Square Error; NWP, Numerical Weather Prediction; PCA, Principal Component Analysis; PID, Proportional-Integral-Derivative; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-analyses; PSO, Particle Swarm Optimization; QDA, Quadratic Discriminant Analysis; RANS, Reynolds-Averaged Navier–Stokes; RBFNN, Radial Basis Function Neural Network; RBM, Restricted Boltzmann Machine; RC, Reactive Control; RES, Renewable Energy Sources; REWS, Rotor-Equivalent Wind Speed; RF, Random Forest; RL, Reinforcement Learning; RLS, Recursive Least Squares; RMSE, Root Mean Square Error; RNN, Recurrent Neural Network; RQ, Research Questions; RT2FNN, Recurrent Type-II Fuzzy Neural Network; SHAP, SHapley Additive exPlanations; SCADA, Supervisory Control and Data Acquisition System; SOC, State of Charge; SOM, Self-Organizing Map; SSA, Singular Spectrum Analysis; SVD, Singular Value Decomposition; SVC, Static Variational Autoencoder Compensator; SVM, Support Vector Machine; SVR, Support Vector Regression; TCN, Temporal Convolutional Network; TLBO, Teaching-Learning Based Optimization; t-SNE, t-distributed Stochastic Neighbor Embedding; TD3, Twin Delayed Deep Deterministic Policy Gradient; UAVs, Unmanned Aerial Vehicles; VMD, Variational Mode Decomposition; WAFs, Windage Alteration Faults; WECs, Wave Energy Converters; WKNN, Weighted k-Nearest Neighbor Regression; WOA, Whale Optimization Algorithm; WPRES, Wind Power Ramps Events; WT, Wind Turbine; XGBR, Extreme Gradient Boosting Regression.

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Multi-objective optimization
Smart control
Fault detection

maintenance, and diagnostics. This systematic review provides a comprehensive evaluation of how machine intelligence has contributed to the optimization of wind energy systems. These techniques have been applied to enhance turbine-level performance, reduce power losses, predict faults, and maximize energy yield under uncertain and dynamic conditions. Particular emphasis is placed on hybrid models that combine data-driven algorithms with physical dynamics and domain heuristics, enabling real-time, predictive, and autonomous wind farm operations. Furthermore, the study critically examines integration barriers such as noisy SCADA data, regulatory compliance, computational costs, and sustainability trade-offs. The findings highlight that multi-objective optimization—balancing energy production, system resilience, and cost efficiency—is central to the most successful implementations. Hybrid frameworks, explainable artificial intelligence (AI), edge computing, and transfer learning are identified as key enablers for scalable deployment. This review offers a comprehensive roadmap for the application of machine intelligence in advancing wind energy optimization and provides actionable insights for researchers, engineers, and policymakers committed to developing intelligent, adaptive, and sustainable wind power infrastructures.

1. Introduction

Wind energy has an efficient power generation method, being the major source of renewable energy responsible for providing a worldwide electricity supply (Yang et al., 2020; Yang et al., 2019) while reducing pollution levels and reliance on non-renewable sources of energy (Song et al., 2020; Li et al., 2019; Chen et al., 2021). The utilization of wind energy has grown significantly in recent years, driven by the need for sustainable energy and the continued decline in the cost of wind energy technologies. The development of larger and more efficient Wind Turbines (WTs) is a significant trend in the wind energy sector (Kamani and Ardehali, 2023). Wind energy also comes with several technical and methodological challenges, and it faces several major problems, including irregular performance due to reductions in wind speed that reduce the power generated (Uz and Chim, 2022; Brennan and van Rensburg, 2023), increase the balance to the supply and demand of electricity and limit its availability as a primary source of electricity (Tan et al., 2022; Deng et al., 2022). The design and performance of WTs also present a major challenge as they must operate in a wide range of wind speeds and directions while remaining resistant to fatigue and other forms of wear and tear. A crucial aspect of wind energy projects is wind resource assessment, which involves measuring wind speeds and directions at different heights to determine potential energy yield (Ala et al., 2023; Al-Quraan et al., 2023). However, this process can be difficult due to the variability of wind speeds, complex terrain or adverse atmospheric conditions in some locations, and therefore, accurate prediction of wind energy is essential for maximizing its utilization. Sophisticated control systems are necessary to allow WTs to respond to changes in wind conditions and maintain stable operation (Orzan et al., 2023; Verma et al., 2023). The integration of wind energy into the electrical grid must be done to ensure a stable and reliable power supply. Wind energy cannot be stored easily, although this problem is being addressed through the development of energy storage technologies, such as batteries and pumped hydro storage (Kong et al., 2022; Tchertchian and Millet, 2023).

The cost of wind energy has been constantly falling, driven by technological advancements and economies of scale. WTs are complex machines that require regular maintenance, downtimes and repairs with elevated operation costs and are difficult to access in remote or offshore locations. Wind energy projects also require a significant upfront investment, and careful planning and management are essential to ensure they are completed on time and within budget (García Marquez et al., 2012). Machine intelligence approaches can play a crucial role in optimizing the performance and efficiency of wind energy systems. These algorithms can analyse large amounts of data on wind speed and direction, as well as the performance of WTs, to identify patterns and trends that can be used to improve the design, placement and operation of WTs (Schwegmann et al., 2023; Adnan et al., 2021). Machine Learning (ML) can help to identify potential problems with WTs before they occur, optimize the operation of WTs in real-time and reduce downtimes and maintenance costs by developing more accurate wind

resource assessments and predictive maintenance models. These algorithms can adjust the pitch of the WT blades, the orientation and other parameters to maximize energy production by analysing data from sensors and other sources. This can improve the overall efficiency and performance of wind energy systems and reach competitiveness in the global energy market (Malakouti et al., 2022; Dubey et al., 2021).

The main objective of this paper is to conduct a comprehensive, systematic review and analysis of novel improvements in machine intelligence approaches to enhance the utilization of wind energy. Several processes, strategies and different computational techniques were analysed to be applied to wind energy improvement. This paper aims to provide valuable insights and recommendations that can be used to develop a decision support system for wind energy deployment. The first objective is to conduct a comprehensive review of the existing literature on artificial intelligence (AI) approaches to optimize wind energy utilization and highlight potential areas for future work. The second objective is to evaluate and compare the performance of different AI approaches to determine their suitability for different wind energy projects. This article serves as a valuable resource for researchers and practitioners working in this field to identify new research lines. Most of the reviews found in the current state of the art are focused on specific applications of ML in wind energy systems, e.g., forecasting of wind power generation or condition monitoring, but without combining these fields. Other reviews are only focused on the bibliometric perspective, analysing the number of cites or keywords without studying the overall overview of ML in current wind energy systems (Zhang et al., 2022; Lagos et al., 2022). This paper provides several contributions to the field of wind energy utilization through AI approaches, including:

- A comprehensive review of the various methods used to enhance the utilization of wind energy through AI approaches. It analyses research published over the past seven years, highlighting the key problems and solutions in this field. Additionally, it discusses recent works by researchers and identifies unanalysed problems that can guide future research in this area.
- This study aims to provide an in-depth analysis of the current research on AI techniques for wind energy advancement. This approach will enable new directions for future study as well as a better understanding of the possible advantages and difficulties for scientific professionals and practitioners.
- The complex nature of wind energy systems and the variability of wind patterns pose significant challenges. Therefore, detailed discussions on how AI approaches can enhance efficiency and reliability serve as the study motivation. This study highlights the key methods and techniques used to address these challenges through a systematic analysis of recent research that discusses the use of ML algorithms, control systems and optimization techniques to enhance the performance of wind energy systems. This analysis identifies the key problems that have not been adequately addressed in the existing literature and suggests future research.

This paper is structured as follows: In [Section 2](#), a set of research questions related to the use of AI approaches in wind energy utilization is presented. The article selection process and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) strategy ([Page et al., 2022](#)) adopted for systematic reviews and meta-analyses were described in [Section 3](#). [Section 4](#) provides a systematic review of the AI methods, e.g., ML and Deep Learning (DL), used in wind energy utilization, including an analysis of the experimental parameters used for accurate calculation and result analysis. In [Section 5](#), observations, limitations, and future research directions are discussed. Finally, conclusions are presented in [Section 6](#).

2. Systematic review and mapping methodology

The purpose of conducting an analytical study is to comprehensively review various methods used for enhancing wind energy utilization through AI approaches. This includes examining the motivations for their use, the techniques employed, the experimental protocols used for validation, and the research obstacles encountered during these investigations. As a result of these considerations, the following Mapping Questions (MQ) and Research Questions (RQ) were developed. The systematic mapping and reviews aim to answer the following questions.

- **MQ1:** What is the current publication status of research on enhancing wind energy utilization with machine intelligence approaches?
- **MQ2:** What machine intelligence approaches are effective for enhancing wind energy utilization?
- **MQ3:** How has the role of machine intelligence approaches in wind energy evolved over time, and what are the current trends in this field?
- **RQ1:** What are the current state-of-the-art machine intelligence approaches for enhancing wind energy utilization?
- **RQ2:** How can ML techniques be used to optimize WT performance and increase energy production?
- **RQ3:** What are the main challenges and limitations associated with the integration of machine intelligence approaches in wind energy systems?
- **RQ4:** What are the potential benefits and drawbacks of using machine intelligence approaches in wind energy utilization compared to traditional approaches?

Search queries were conducted on digital libraries such as IEEE Explore, Springer, Science Direct, Wiley, Nature, IET, Sage, Taylor & Francis and MDPI to find studies on DL, ML and other relevant methodologies. The search results were reviewed using titles and authors to ensure no duplicates, selecting 975 relevant papers. The content and methods of the papers were analysed thoroughly, being compared with a set of inclusion and exclusion criteria shown in [Table 1](#). These inclusion

criteria aim to identify relevant papers that provide valuable insights into the application of machine intelligence approaches in the field of wind energy. The exclusion criteria help to maintain the focus and validity of the study and minimize the risk of including irrelevant papers.

3. Systematic mapping report

3.1. MQ1: What is the current publication status of research on enhancing wind energy utilization with machine intelligence approaches?

[Table 2](#) shows the number of papers selected each year for the past seven years that highlight the role of machine intelligence approaches in wind energy. Additionally, it shows the percentage utilization of papers reviewed each year.

A total of 169 papers were selected for study and analysis based on the selection criteria mentioned in [Table 1](#). The distribution of these papers across different publishers is shown in [Fig. 1](#).

[Fig. 2](#) depicts the flowchart for the PRISMA-style approach used for article selection in this study. In the initial search and screening phase, the search query was used to retrieve articles from databases using relevant keywords and the articles were filtered based on publication type and the time frame of the study. The keywords combinations are “wind energy”, “machine learning”, “wind turbine optimization”, “artificial intelligence”, “wind power prediction”, “deep learning”, “Wind resource assessment”, “data-driven models”, “wind energy forecasting”, “machine learning”, “Wind turbine fault detection”, “machine learning”, “Wind energy integration”, “intelligent control systems”, “predictive maintenance”, “wind turbines”, “ML or AI” and “Energy systems”, “machine intelligence” and “wind energy”.

Only journal publications and conference papers from the last seven years (2018–2024) were included because the research trend showed a high contribution towards AI approaches in this time frame. Book chapters, abstract-only papers, and editorial and poster papers were excluded, and 69 duplicate papers were also excluded, leaving 975 papers for the study. In the full-text assessment and exclusion criteria phase, the full text of the remaining 975 papers was assessed based on inclusion and exclusion criteria. The criteria included methods, outcomes, comparative study, inclusion and exclusion criteria, discussion and challenges. For this case, 224 records were excluded considering the abstract, study relevancy and title. After content analysis based on full-text assessment, 408 records were excluded, leaving 343 papers. In the final selection phase, only papers related to wind energy and methodological combination were considered, and 174 records were removed based on this criterion. Finally, 169 articles were chosen based on their relevance to the research and the hierarchical record exclusion method described above.

[Fig. 3](#) outlines a systematic review process for evaluating machine intelligence approaches in wind energy utilization. It begins with research question design and article selection, followed by data synthesis on key themes like ML, DL, data sources, and publication trends. The analysis phase assesses method performance, findings, and challenges, leading to a comprehensive conclusion.

Table 1
Inclusion and exclusion criteria.

S. No	Inclusion Criteria
1	The paper should be focused on exploring or analysing machine intelligence approaches to optimize the performance and efficiency of wind energy systems.
2	The paper should make use of various computational and statistical methods to analyse data or develop models related to wind energy and machine intelligence.
3	The paper must be published in a peer-reviewed journal or conference proceedings.
	Exclusion Criteria
1	Language: Papers written in other languages will be excluded due to language barriers.
2	Accessibility: Papers that are not available in electronic format will be excluded.

Table 2
Year-wise classification of selected papers by publication type.

S. No.	Year of Publication	Number count	(%)
1	2024	24	14.20
2	2023	27	15.97
3	2022	29	17.15
4	2021	18	10.65
5	2020	25	14.79
6	2019	23	13.60
7	2018	23	13.60
	Total	169	100

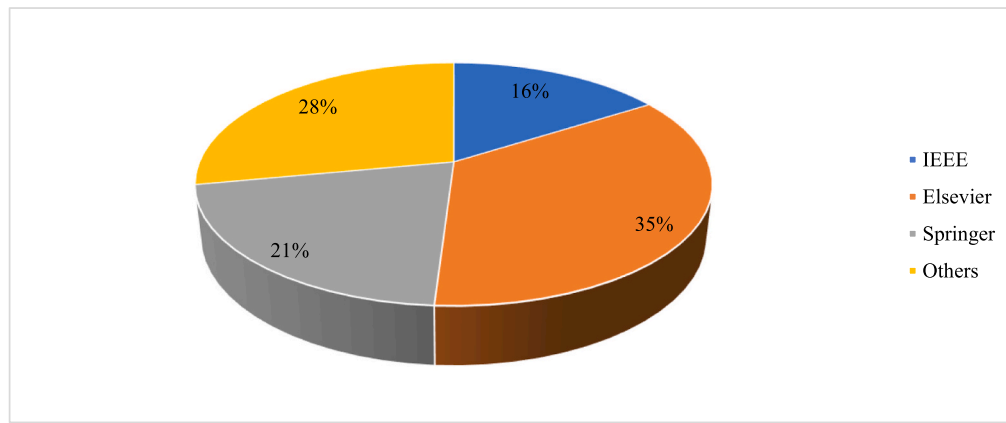


Fig. 1. Paper distribution from different publishers according to the inclusion and exclusion criteria.

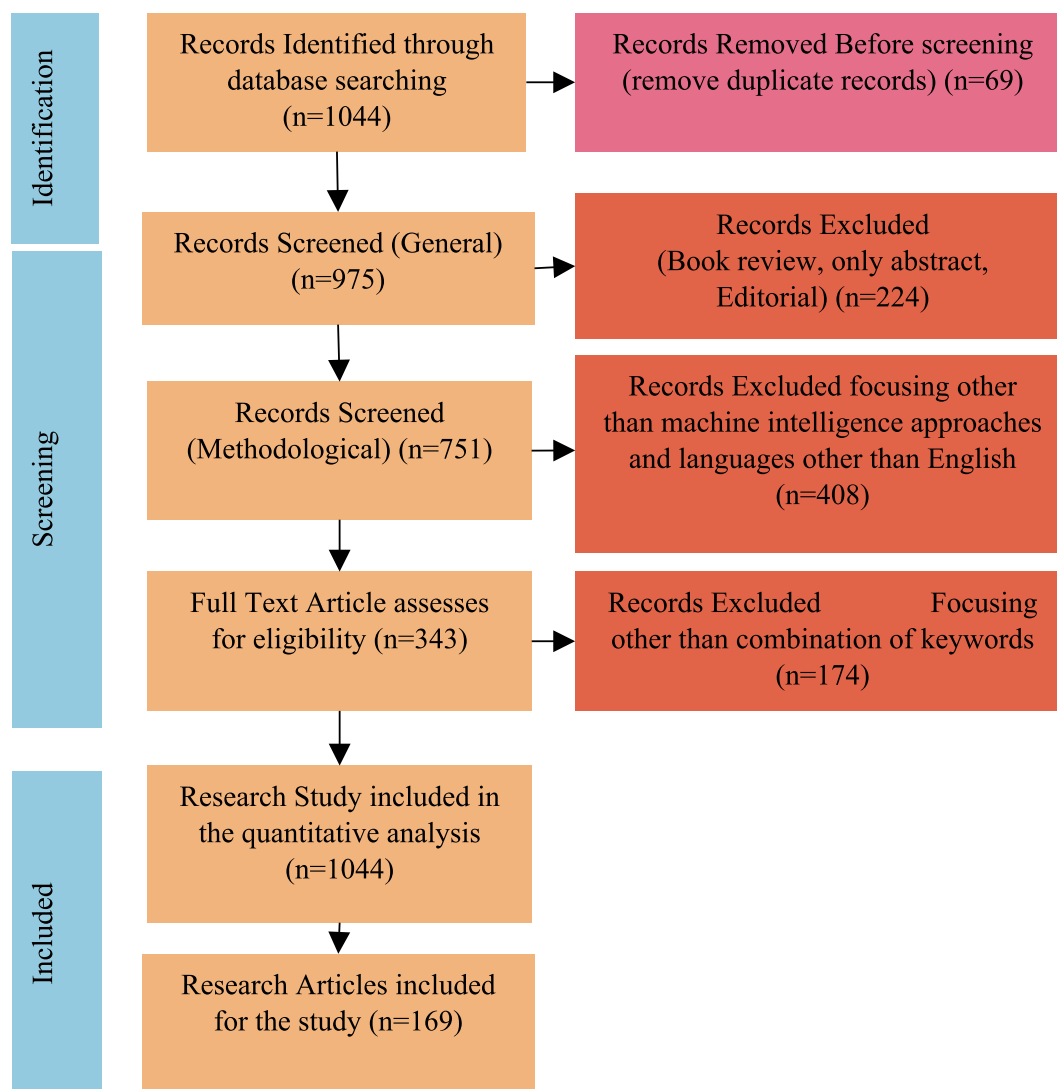


Fig. 2. Flowchart for the PRISMA Style (Article Selection Strategy).

3.2. MQ2: What machine intelligence approaches are effective for enhancing wind energy utilization?

Wind energy is a promising renewable energy source that can help reduce greenhouse gas emissions and combat climate change. However,

the efficiency and reliability of wind energy generation systems are still limited by various factors, such as wind speed, wind direction, turbulence, and intermittency, among others. Machine intelligence approaches have been increasingly applied in recent years to overcome these challenges and improve wind energy utilization. Several major

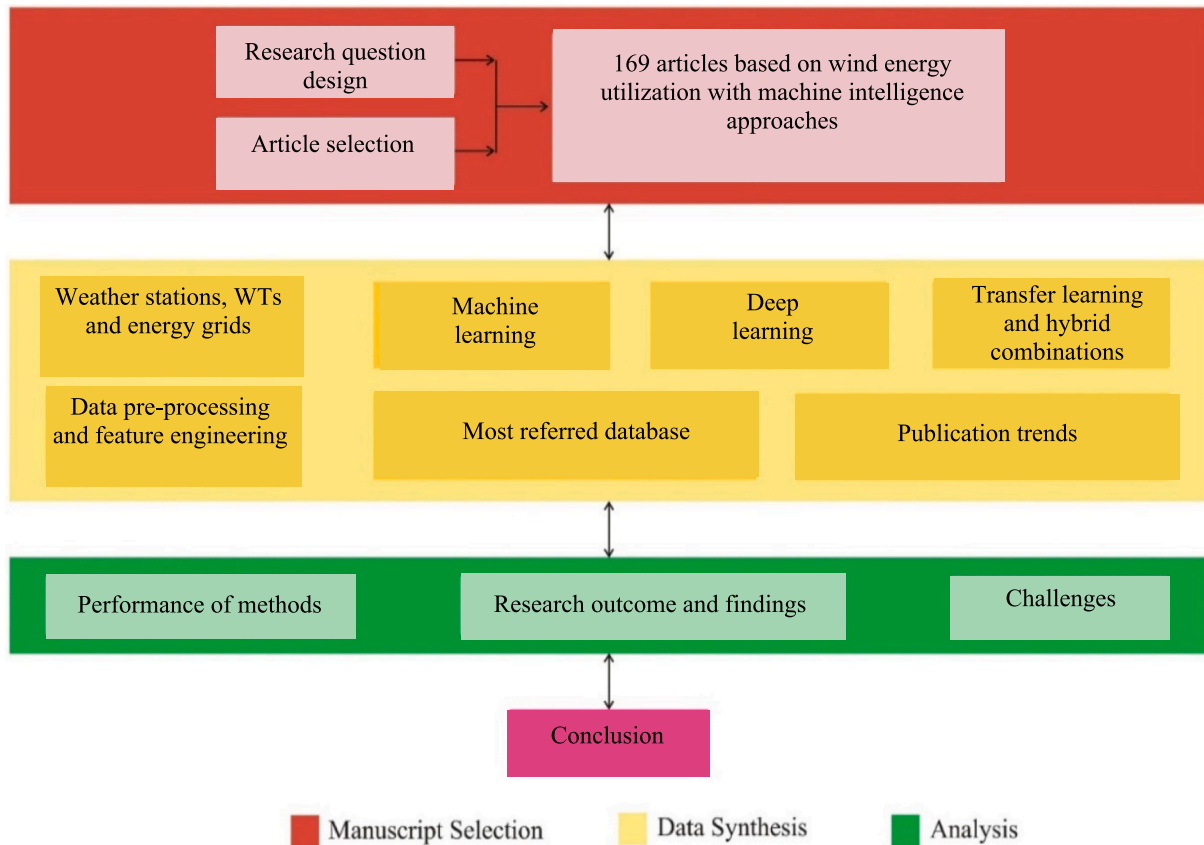


Fig. 3. Manuscript selection strategy, data synthesis and analysis.

methodological categories were considered for the review and analysis of enhancing wind energy utilization using machine intelligence approaches.

Predictive modelling

One of the most promising applications of AI in wind energy is the use of predictive models to optimize WT performance (Luo et al., 2018; Zhang and Chen, 2022). They can use historical data and real-time data to forecast wind speed and direction and adjust the operation of WTs. WTs can operate at maximum efficiency and generate more electricity by adjusting the blade pitch angle, rotor speed and other parameters based on real-time data. Different ML algorithms, e.g., Artificial Neural Network (ANN), Support Vector Machines (SVM) and Random Forest (RF), can be used to build predictive models.

Smart control systems

Another approach for enhancing wind energy utilization using machine intelligence is the use of smart control systems that can use ML algorithms to continuously monitor wind conditions and adjust WT operation accordingly (Ali et al., 2022; Azeroual et al., 2022; Sierra-Garcia et al., 2022; Daraz et al., 2022). A relevant application is based on the detection of sudden changes in wind speed or direction and adjusting the blade pitch angle and rotor speed to maintain constant power output. Smart control systems can also detect and mitigate the effects of turbulence and wind shear, reducing the efficiency and reliability of WTs. Smart control systems can improve energy efficiency, reduce maintenance costs and extend the lifespan of WTs through the application of AI approaches to optimize WT operation (Sierra-Garcia et al., 2022; Daraz et al., 2022; Dassa and Recioui, 2022; Bedi et al., 2022; Amir et al., 2022).

Wind farm layout optimization

AI can also be used to optimize wind farm layouts (Cazzaro and Pisinger, 2022; Zhong et al., 2024). Wind farm layouts can affect the efficiency and performance of wind energy generation systems. Wind

farm layouts can be optimized for maximum energy production through ML algorithms that analyse wind speed and direction data, as well as other environmental factors, e.g., ML algorithms can identify areas with the highest wind speeds and adjust the placement of WTs accordingly (Thomas et al., 2022; Hidayat et al., 2022). The optimization of wind farm layouts and wind energy generation systems achieves higher energy production and reduces the cost of energy.

Fault detection and maintenance

Another application of these techniques in wind energy is the application of data analytics to monitor WT performance and identify potential issues (Dubey et al., 2022). The collection and analysis of data from sensors and other sources allow data analytics that can detect anomalies in WT performance, e.g., changes in vibration, temperature and power output. Maintenance operators can perform corrective tasks before these issues become more severe by the early identification of these issues (Nayl et al., 2022). Data analytics can also be used to optimize maintenance schedules and reduce downtime, which can further improve the efficiency and reliability of wind energy generation systems.

Weather forecasting

This methodology can also be used to improve weather forecasting, helping to optimize wind energy generation systems. Weather forecasting models can be built to predict wind speed, direction and other relevant parameters with greater accuracy through ML algorithms that analyse historical weather data, satellite images and other sources (Nayl et al., 2022; Hess and Boers, 2022; Singh et al., 2022). These predictive models can then be used to optimize WT operation and improve energy efficiency.

3.3. MQ3: How has the role of machine intelligence approaches in wind energy evolved over time, and what are the current trends in this field?

Advances in energy storage technology, such as batteries and pumped hydro storage, are helping to address the challenge of grid integration, allowing wind energy to provide a more stable source of electricity. Currently, the trend in this field is towards the use of advanced machine intelligence approaches, e.g., DL and Reinforcement Learning (RL), for improved WT control, operation and maintenance (Zhang and Chen, 2022; Zhang and Li, 2022). Additionally, there is a growing interest in the integration of ML with other technologies, such as the Internet of Things (IoT) and blockchain, for real-time monitoring and control of wind energy systems (Dubey et al., 2021; Azeroual et al., 2022; Wu et al., 2022; Howland et al., 2022; Yang et al., 2022). Another trend is the application of machine intelligence approaches to address the challenges of wind energy integration into the grid, such as grid stability and power quality issues (Dubey et al., 2021; Zhang and Chen, 2022; Wu et al., 2022; Howland et al., 2022; Zhang et al., 2021; Shao et al., 2023). Finally, there is an increased focus on the use of ML for wind energy forecasting, which is crucial for efficient grid integration and optimal operation of wind energy systems. Fig. 4 shows the timeline chart of the various contributions to wind energy utilization with machine intelligence approaches. It has been selected based on the relevancy, usage and citations considering current trends. Fig. 5 shows the complete keyword analysis from the selected papers.

4. Systematic review report based on machine-intelligent methods

ML techniques applied in wind energy systems are broadly categorized into supervised learning, unsupervised learning, and RL (Nemade et al., 2022) (Fig. 6).

Taxonomy of machine intelligence approaches in wind energy systems

Supervised learning uses labelled data to address regression and classification tasks. Regression methods such as Linear Regression (LR), logistic regression, polynomial regression, ridge regression, and Least

Absolute Shrinkage and Selection Operator (LASSO) are commonly used for wind speed forecasting, power estimation, and time-series analysis (Wu and Liu, 2021; Keshmiry et al., 2023). Classification algorithms like Naive Bayes (NB), SVM, Decision Tree (DT), ANN, K-Nearest Neighbor (KNN), and RF are widely applied for fault detection, turbine health assessment, and anomaly classification (Tyralis et al., 2021).

Unsupervised learning extracts hidden structures from unlabelled data using techniques such as dimensionality reduction and clustering (Keshmiry et al., 2023; Hassan and Rashid, 2021). Methods including Principal Component Analysis (PCA), t-distributed Stochastic Neighbor Embedding (t-SNE), Linear Discriminant Analysis (LDA), and Singular Value Decomposition (SVD) are effective for simplifying complex Supervisory Control and Data Acquisition System (SCADA) datasets and enabling meaningful feature extraction. Clustering techniques such as K-Means clustering, fuzzy clustering, hierarchical clustering, and distribution-based clustering, support condition monitoring and operational state grouping (Adnan et al., 2021; Nemade et al., 2022; Tyralis et al., 2021). Additionally, generative models like autoencoders, variational autoencoders, and Generative Adversarial Networks (GAN) are used for data reconstruction, augmentation, and noise filtering in DL applications (Dalal et al., 2023).

RL involves agents learning optimal actions through environmental interaction and reward feedback. It is increasingly used in WT control, particularly for dynamic pitch and yaw regulation. Enhancements to RL, such as supervised pretraining, transfer learning, and unsupervised pretraining, improve learning efficiency (Adnan et al., 2021; Nemade et al., 2022; Wu and Liu, 2021; Tyralis et al., 2021; Hassan and Rashid, 2021). Deep models, including Convolutional Neural Network (CNN), Restricted Boltzmann Machines (RBM), Deep Belief Network (DBN), Recurrent Neural Network (RNN), and Self-Organizing Maps (SOM), further support real-time control and intelligent decision-making under varying operational conditions (Adnan et al., 2021; Nemade et al., 2022; Wu and Liu, 2021; Tyralis et al., 2021; Hassan and Rashid, 2021).

Table 3 presents a structured taxonomy of machine intelligence approaches based on learning paradigms (category), model types (models/approaches/techniques), application domains, and associated data inputs commonly used in wind energy systems. The review and analysis in

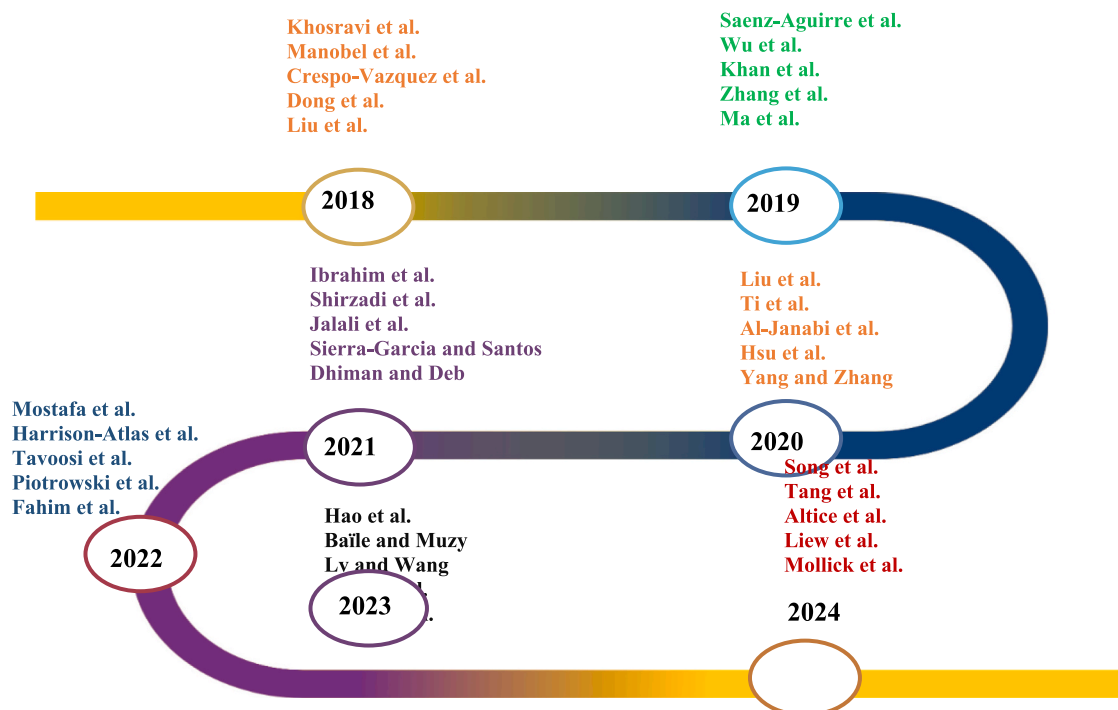


Fig. 4. A timeline chart of the various contributions in wind energy utilization with machine intelligence approaches.

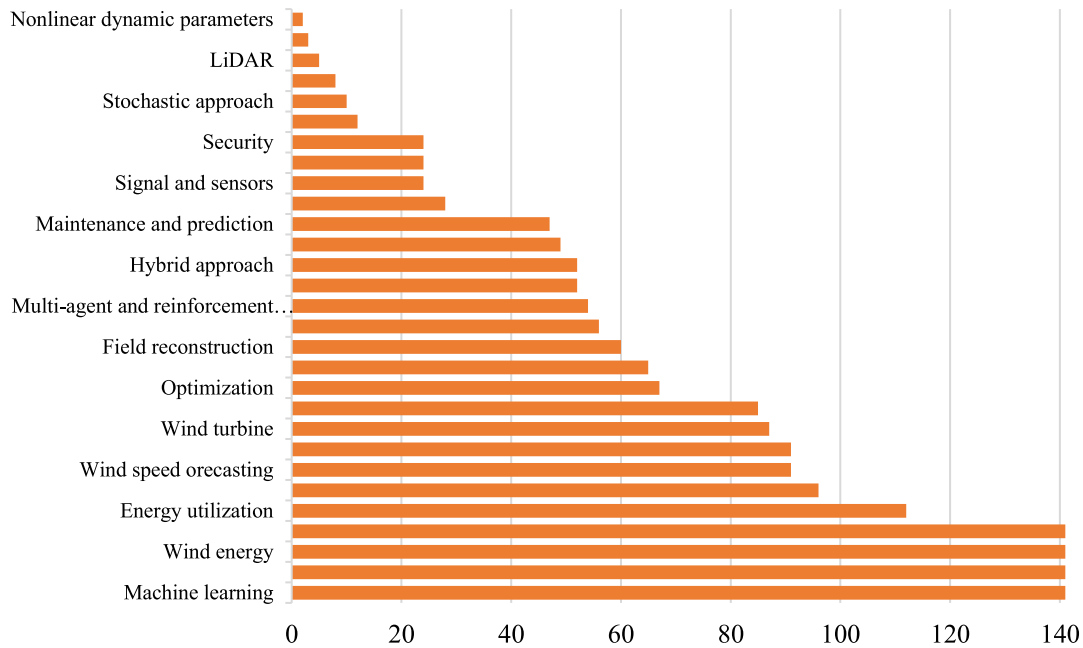


Fig. 5. Complete keyword analysis from the selected papers.

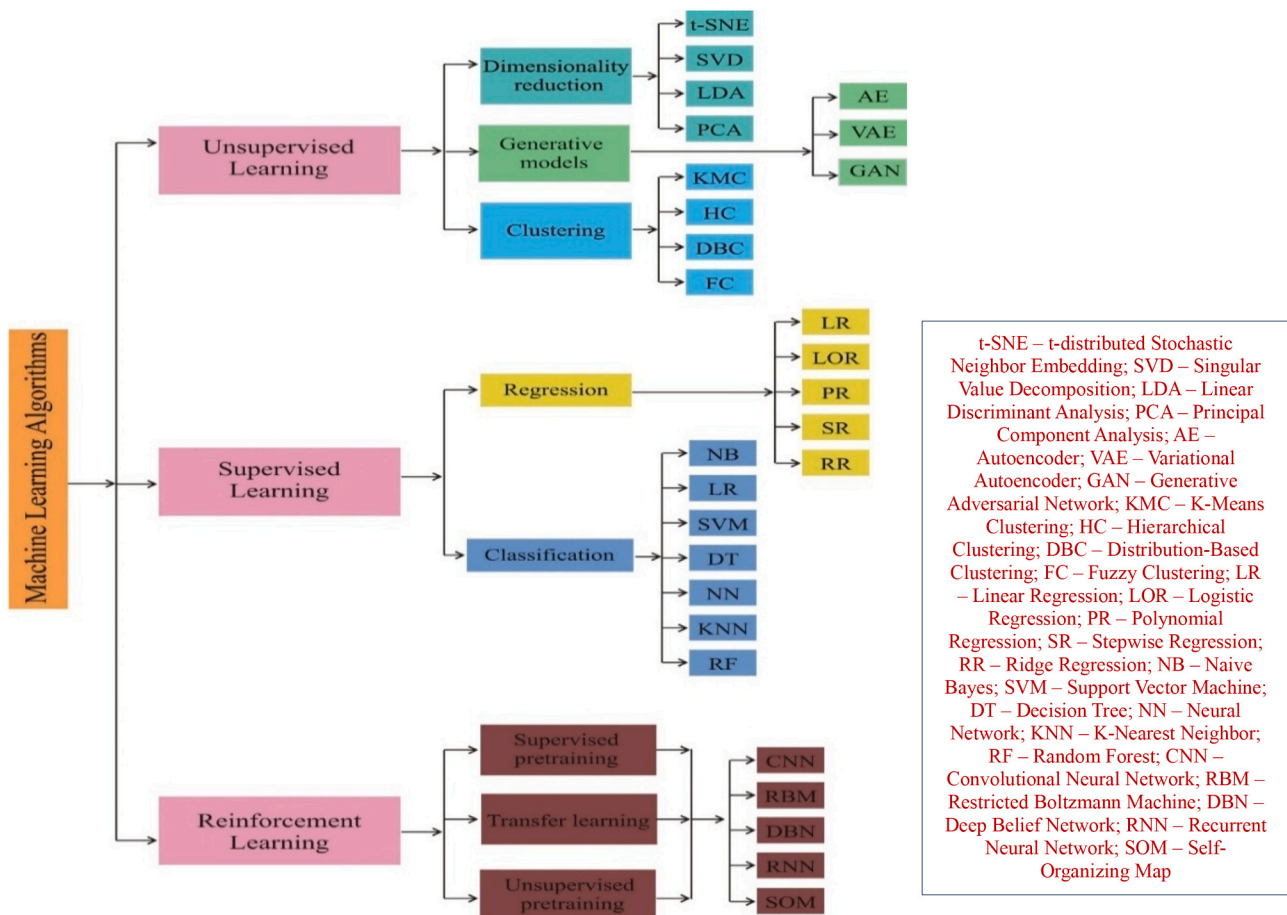


Fig. 6. Taxonomic structure of machine learning algorithms applied in wind energy systems.

this paper are based on these categories.

This taxonomy used in this paper classifies machine intelligence methods applied in wind energy systems into seven principal categories:

supervised learning, unsupervised learning, RL, DL, hybrid models, optimization techniques, and physics-informed ML. Each category is distinguished by its learning paradigm, application domain, and data

Table 3

Taxonomy of machine intelligence approaches in wind energy systems based on model type, application domain, and data input.

S. No	Category	Models/Approaches/Techniques	Applications	Data types	Characteristics
1	Supervised Learning	SVM, RF, ANN, KNN, DT, SVR, MLP	Wind speed and power forecasting, fault detection, performance estimation	SCADA, NWP, LiDAR	Widely used for regression and classification; interpretable; prone to overfitting on noisy data.
2	Unsupervised Learning	K-means, DBSCAN, PCA, t-SNE, Autoencoders, GANs	Clustering, anomaly detection, feature extraction, data denoising	SCADA, operational logs, image data	Useful for pre-processing and pattern discovery; no label requirement
3	Reinforcement Learning	Q-learning, DQN, Multi-Agent DRL, DDPG	Control strategies, yaw/pitch optimization, battery scheduling	Sensor data, control loop sequences	Learns policies over time; suited for real-time decision-making; sample-efficient variants emerging
4	Deep Learning	CNN, LSTM, GRU, Transformer, DBN, Deep Autoencoders	Spatiotemporal forecasting, fault classification, blade inspection	SCADA, drone imagery, meteorological data	Handles high-dimensional, nonlinear data; needs large datasets and high computation
5	Hybrid Models	CNN-GRU, ANFIS-PSO, PCA-DL, VMD-LSTM, ANFIS-GA, LSTM-GRU, EWT-LSTM	Forecasting, ramp event detection, predictive maintenance	SCADA + NWP + weather inputs	Combine strengths of multiple methods; usually highest accuracy; complex to train and tune
6	Optimization Techniques	PSO, GA, GWO, WOA, TLBO	Hyperparameter tuning, control optimization, feature selection	Acts on model or system parameters	Enhance performance of other models; not predictive alone
7	Physics-informed ML	CFD-ML hybrids, RANS-ML models, Wake flow integration	Offshore wind modeling, wake prediction, flow field estimation	CFD outputs, wind tunnel and site data	Bridges physical knowledge and data-driven learning; interpretability retained but less mature

handling capability. Supervised models are predominantly used for wind speed forecasting, power prediction, and fault detection due to their reliance on labeled data. Unsupervised methods are valuable for feature extraction and anomaly detection in SCADA datasets, especially under noisy or unlabeled conditions. RL and hybrid models are typically deployed in dynamic control and energy optimization settings, where adaptability and learning from interaction are crucial. Optimization techniques such as GA and PSO enhance model performance by tuning hyperparameters and improving search strategies. Physics-informed models represent an emerging class that bridges empirical ML with physical flow dynamics, particularly in offshore or complex terrain scenarios. This structured taxonomy supports the analytical organization of the subsequent review sections covering RQ1 to RQ4.

4.1. RQ1: What are the current state-of-the-art machine intelligence approaches for enhancing wind energy utilization?

This section primarily analyzes supervised learning, DL, and hybrid models commonly used for forecasting, control, and performance optimization in wind energy systems.

4.1.1. Techniques for wind energy forecasting and optimization

Recent advancements in AI have significantly improved wind energy forecasting and system optimization. Forecasting wind speed and direction is critical for maximizing turbine efficiency, enhancing grid reliability, and ensuring energy market stability. Time-series prediction techniques, often leveraging historical data, are widely employed to anticipate wind behavior and support real-time operational decisions. ML-based models also facilitate knowledge transfer across different wind farm sites and geographical locations, enabling more adaptive forecasting systems. Power optimization, another vital application, benefits from algorithms that intelligently manage generation and distribution to align with grid demands. These data-driven methods support the integration of wind energy into broader energy systems, particularly in coordination with energy storage solutions. Overall, the application of ML in forecasting and power optimization enhances operational efficiency, minimizes curtailment, and supports informed maintenance planning.

4.1.2. Machine learning applications in Control, Reliability, and power modeling

Advanced ML techniques are increasingly employed for control, reliability, and power modeling in WTs. For instance, Jiménez et al. (Jiménez et al., 2020) demonstrated the superiority of a Nonlinear Autoregressive Exogenous (NARX) model over linear autoregressive

models in detecting blade delamination. Dong et al. (Dong et al., 1975) developed a hybrid approach combining phase space reconstruction, Adaptive Neuro-Fuzzy Inference System (ANFIS), and Particle Swarm Optimization (PSO) for performance modeling. Vijayshankar et al. (Vijayshankar et al., 2021) proposed a Deep Reinforcement Learning (DRL)-based real-time control system for tracking power reference signals, incorporating wake effects and dynamic atmospheric data. Tavoosi et al. (Tavoosi et al., 2022) utilized a Recurrent Type-II Fuzzy Neural Network (RT2FNN) to regulate power and voltage in turbines using Doubly Fed Induction Generators (DFIGs), achieving improved power quality. Addressing market-side uncertainty, Crespo-Vazquez et al. (Crespo-Vazquez et al., 2018) developed a decision-making framework combining multivariate clustering and RNN to model fluctuations in energy prices and availability. Collectively, these methods demonstrate the potential of ML in enhancing real-time responsiveness, fault resilience, and energy price-informed operations in wind energy systems.

4.1.3. Artificial intelligence and hybrid forecasting models for Real-World implementations

Wind power forecasting has emerged as a dominant area of research in machine intelligence applications. Recent works have focused on hybrid and DL models to improve forecast accuracy, capture uncertainty, and enhance real-time operational capabilities. Wu et al. (Wu et al., 2019) utilized stacked denoising autoencoders with Long Short-Term Memory (LSTM) networks for deep feature extraction, while Hong and Rioflorido (Hong and Rioflorido, 2019) integrated CNN with Radial Basis Function Neural Network (RBFNN) for 24-hour forecasting. Khan et al. (Khan et al., 2019) introduced a hybrid model based on PCA and DL to remove feature correlation and improve wind prediction.

Probabilistic and ensemble forecasting approaches have also gained attention. Sun et al. (Sun et al., 2019) proposed a two-step framework using Support Vector Regression (SVR) for pinball loss optimization, while Luo et al. (Luo et al., 2018) employed Extreme Learning Machine (ELM) for handling large-scale datasets. Liu et al. (Liu et al., 2020) integrated empirical wavelet transform (EWT) with deep networks and RL for robust wind speed forecasting.

Several studies tackled short-term power and reactive power prediction under uncertainty. Shirzadi et al. (Samet et al., 2020) used CNNs to forecast reactive power and mitigate voltage flicker through a Static Variational Autoencoder Compensator (SVC), while Piotrowski et al. (Piotrowski et al., 2022) incorporated Numerical Weather Prediction (NWP) for large-scale wind farms using ensemble averaging. Mishra et al. (Mishra et al., 2020) benchmarked five DL architectures for multivariate forecasting using pre-processed signal data. Similarly, Salah et al. (Salah et al., 2022) and Zhu et al. (Zhu et al., 2018) demonstrated

improvements in prediction accuracy using ensemble methods and spatiotemporal correlation, respectively.

Hybrid ML models incorporating optimization and feature selection techniques have shown promise. Khosravi et al. (Khosravi et al., 2018) used ANFIS with PSO and Genetic Algorithm (GA). Liu et al. (Liu et al., 2018) and Yi et al. (Yi et al., 2019) combined LSTM with empirical decomposition or evolutionary learning for wind speed forecasting. Santhosh et al. (Santhosh et al., 2019) and Saini et al. (Saini et al., 2020) used dynamic RNNs integrated with NARX and extreme learning models for smart grid applications.

The forecasting of ramp events—sudden fluctuations in power—was addressed using multi-task learning to predict Wind Power Ramps Events (WPRES) (Dorado-Moreno et al., 2020), CNN-twin SVR hybrid models (Dhiman and Deb, 2021), and Deep RL frameworks (Futakuchi et al., 2021). These approaches improved stability and reduced ramping frequency. Jeong and Kim (Jeong and Kim, 2021) proposed DeepComp, a dispatch error compensation framework using RL, while the study in (Rashid et al., 2020) presents a RF regressor for wind power prediction using SCADA data, achieving low errors across capacity factors; however, the lack of comparison with other models and limited statistical validation weakens its generalizability and benchmarking strength.

Emerging studies also explored physical system modeling and explainability. Richmond et al. (Richmond et al., 2020) proposed a Computational Fluid Dynamics (CFD)-integrated ML approach for offshore flow field prediction. Mohandes et al. (Mohandes and Rehman, 2018) used Deep Neural Network (DNN) with Light Detection and Ranging (LiDAR) data for height extrapolation. Rushdi et al. (Rushdi et al., 2020) demonstrated ML-based force prediction in airborne kite systems. Bui et al. (Bui et al., 2020) introduced a cooperative multi-agent DRL system to optimize wind farm output by minimizing wake effects.

Overall, these hybrid models offer improvements in accuracy, uncertainty handling, and generalizability. However, challenges remain in model interpretability, large-scale data processing, and real-world deployment under varying operational and climatic conditions.

4.2. RQ2: How can ML techniques be used to optimize WT performance and increase energy production?

This section focuses on how supervised and DL techniques, particularly CNN, LSTM, and hybrid architectures have been applied to enhance WT performance, fault detection, and power estimation.

This section provides a detailed discussion with a primary focus on WTs, covering various aspects, e.g., damage inspection, estimation of power outputs, status assessment, automated drivetrain fault detection, blade icing fault detection, yaw misalignments, power curve modelling, data imputation, anomaly detection, etc.

ML techniques have significantly advanced the performance optimization of WTs through enhanced condition monitoring, predictive maintenance, and adaptive control strategies. Condition monitoring systems leveraging DL architectures such as CNN and ELM have enabled real-time fault recognition, particularly for drivetrain anomalies (Bach-Andersen et al., 2018; Wang et al., 2023). Integrated frameworks proposed by Vives et al. (Vives et al., 2020), Shihavuddin et al. (Shihavuddin et al., 2019), and Moreno et al. (Moreno et al., 2018) demonstrated effective early fault detection through automated learning, drone-based blade inspections, and damage assessment in WT blades, respectively.

Acoustic analysis using ML models has further supported fault identification. Studies by Ciaburro et al. (Ciaburro et al., 2021) and García Márquez et al. (García Márquez et al., 2021) utilized signal classification and wavelet transforms to isolate WT noise signatures. For blade icing, SCADA-driven models incorporating DNN, autoencoders, and climatic variables have proven effective in pre-emptive fault detection (Liu et al., 2018; Chen et al., 2019; Kreutz et al., 2019).

SCADA data has also enabled broader health tracking applications.

Multiple studies proposed ML-based frameworks to predict faults, assess maintenance needs, and estimate wind direction using LiDAR (Choi et al., 2018; Zhao et al., 2018; Liu et al., 2022; Teng et al., 2018; Yeh et al., 2019). Gearbox monitoring remains a critical focus due to its high failure rate. Solutions integrating Joint Variational Autoencoder (JVAE), regression models, and deep belief networks have shown strong predictive performance in forecasting failures (Yang et al., 2020; Koukoura et al., 2019; Rashid et al., 2020; Lu et al., 2019).

False alarm reduction is another critical domain. Ensemble classifiers and semi-supervised approaches have improved classification accuracy while reducing unnecessary maintenance responses (Peco Chacón et al., 2022; Peco Chacón et al., 2023; Jiang et al., 2022). To mitigate the impact of missing or noisy data, researchers have employed GAN and statistical process control methods, improving both imputation accuracy and fault diagnosis (Qureshi and Khan, 2019; Hsu et al., 2020).

In addition to diagnostics, ML models have advanced forecasting and control. Studies demonstrated improvements in yaw alignment, wake modeling, and energy yield forecasting by integrating DL and hybrid models with CFD, fuzzy logic, and RL (Sierra-García et al., 2022; Kulkarni et al., 2019; Ti et al., 2020; Lin and Liu, 2020). Sierra-García and Santos (Sierra-García et al., 2022) developed a fuzzy logic–DL hybrid controller that outperformed both traditional Proportional-Integral-Derivative (PID) and fuzzy controllers. Techniques like Q-learning and PSO were applied to adjust yaw angles in real-time for mechanical load reduction (Saenz-Aguirre et al., 2019).

Power output estimation has been enhanced using models that correlate environmental conditions with generation metrics (Aksoy and Selbaş, 2021; Lin et al., 2021). Power curve modeling has also been refined through techniques such as Gaussian Processes (GP), SVR, and DL-based regression models to detect nonlinear behavior and operating anomalies (Manobel et al., 2018; Pandit and Infield, 2019).

Recent developments have explored fault mechanics in offshore and bladeless turbines, where DL and LSTM-based simulations enabled low-error force modeling and performance forecasting under turbulent flow conditions (Liu et al., 2022; Yin et al., 2021; Dehghan Manshadi et al., 2022; Díaz et al., 2018; Aweda et al., 2022). Díaz et al. (Díaz et al., 2018) assessed long-term power predictions using Measure-Correlate-Predict (MCP) methods with models including ANN, SVM, and RF, emphasizing separate treatment of wind speed and air density. These efforts reflect a shift toward integrated, predictive, and real-time turbine optimization under diverse environmental conditions (Rodríguez et al., 2023) (Salcedo-Sanz et al., 2018) (Fu et al., 2022).

4.3. RQ3: What are the main challenges and limitations associated with the integration of machine intelligence approaches in wind energy systems?

This section critically examines the limitations associated with applying RL, hybrid models, and unsupervised methods under practical deployment constraints.

This section provides a discussion with a primary focus on the integration of machine intelligence approaches in wind energy systems, covering various aspects such as big data analytics, forecasting, knowledge transfer, hybrid decomposition, etc.

Machine intelligence offers substantial benefits to wind energy systems, but its integration introduces several practical and technical challenges. Xu et al. (Xu et al., 2018) highlighted energy consumption issues in data centers and proposed a RL-based job scheduling algorithm enhanced by an ANN to reduce training costs. Sharifzadeh et al. (Sharifzadeh et al., 2019) addressed uncertainty in renewable energy grids, demonstrating that SVR, GP, and ANN can model demand-side fluctuations, with ANN yielding the most effective results.

Predictive maintenance has been explored through data-centric strategies. Garan et al. (Garan et al., 2022) used DT-based models and optimized feature selection to extend the useful life of WTs. Yang et al. (Yang and Zhang, 2020) applied a two-step DL framework using a variational autoencoder and Multi-Layer Perceptron (MLP) to improve

aerodynamic design via inverse optimization. These methods reduce reliance on computationally expensive iterative processes.

Ensemble and hybrid models have been proposed for forecasting under uncertainty. Du (Du, 2018) combined ANN, SVR, and GP for wind power forecasting using meteorological data. Ma et al. (Ma et al., 2019) demonstrated that a model combining wavelet transform, negative correlation learning neural network, and PSO outperformed other ensemble methods for short-term wind speed forecasting. The results were compared with Back-Propagation Neural Network (BPNN), SVM, bagging, AdaBoost, RF, LSTM, seasonal autoregressive integrated moving average and ensemble SVM. Fahim et al. (Fahim et al., 2022) presented a 5G-Next Generation-Radio Access Network (5G-NG-RAN) supported digital twin for remote turbine monitoring, outperforming traditional models in power forecasting.

Big data analytics also plays a vital role. Zhang et al. (Zhang and Li, 2022) utilized extreme learning with hybrid mode decomposition and a crisscross optimization algorithm. Mostafa et al. (Mostafa et al., 2022) developed a framework using penalized LR, CNN, and gradient boosted trees to predict smart grid stability, emphasizing the challenge of limited dataset sizes. Harrison-Atlas et al. (Harrison-Atlas et al., 2022) introduced a CNN-based surrogate model that efficiently predicted Annual Energy Production (AEP), outperforming GP-based models and enabling national-scale performance assessment.

Despite these advances, challenges remain in ensuring model interpretability, infrastructure compatibility, data quality, and deployment scalability. Addressing these gaps is critical for realizing the full value of intelligent wind energy systems in dynamic operational environments.

4.3.1. Evaluative comparison of machine intelligence approaches for wind energy

Numerous studies demonstrate that hybrid and DL-based models are better suited for handling noisy SCADA data and uncertainty in wind energy systems than conventional approaches. Liu et al. (Liu et al., 2018) proposed a combination of Variational Mode Decomposition (VMD), Singular Spectrum Analysis (SSA), LSTM and ELM model that achieved the lowest Mean Absolute Percentage Error (MAPE) (6.49 %) and Root Mean Square Error (RMSE) (0.4038) through effective decomposition and temporal modeling. Likewise, Liu et al. (Liu et al., 2020) integrated EWT with LSTM, DBN, ESN, and Q-learning, achieving 92 % accuracy improvement over baselines in noisy SCADA settings. Santhosh et al. (Santhosh et al., 2019) applied Ensemble Empirical Mode Decomposition (EEMD) with deep Boltzmann machine, showing a 59–87 % gain in accuracy over 11 benchmarks, with RMSE as low as 0.1238 and MAPE at 0.9941. These studies confirm that decomposition methods such as VMD, SSA, EWT, and EEMD significantly enhance noise resilience.

Statistical validation was a central focus in many of these works. Khosravi et al. (Khosravi et al., 2018) demonstrated that ANFIS-PSO and ANFIS-GA outperform standalone fuzzy inference systems under noisy conditions, while Group Method of Data Handling (GMDH) proved most consistent across all intervals. Similarly, Hossain et al. (Hossain et al., 2021) utilized a CNN– Gated Recurrent Unit (GRU) architecture with Harris Hawks Optimization (HHO) to reduce MAPE by 37.8 % (5-min) and 24.1 % (10-min) over traditional models. Nezhad et al. (Nezhad et al., 2021) showed that Generalized Regression Neural Network-Whale Optimization Algorithm (GRNN-WOA) outperformed GRNN-Grey Wolf Optimization (GWO) and Radial Basis Function (RBF)-PSO under seasonal and marine noise variability with RMSE = 0.0205 and MAPE = 6.84 %. Khan et al. (Khan et al., 2019) used PCA with a TensorFlow-based DNN, reducing RMSE to 0.0553 and Mean Absolute Error (MAE) to 0.1503. Piotrowski et al. (Piotrowski et al., 2022) introduced “ensemble averaging without extremes,” achieving the best normalized MAE (12.44 %) and skill scores (up to 0.465) using advanced ensemble models, proving robust across 29 months of NWP-based data.

CNN-based fault and maintenance frameworks also show high resilience to noise. Yeh et al. (Yeh et al., 2019) proposed a CNN-SVM

model achieving 95 % accuracy in long-cycle maintenance forecasting, outperforming rule-based methods using 4.2 million SCADA records. Bach-Andersen et al. (Bach-Andersen et al., 2018) demonstrated that CNNs trained on vibration signals outperformed logistic regression and fuzzy clustering networks in early fault detection, reaching Area Under the Curve (AUC) as 0.996. Deep autoencoders combined with extreme value theory (Zhao et al., 2018) detected turbine faults 8–14 h earlier than traditional models, reducing false positives under fluctuating SCADA inputs.

In control and operational optimization, RL approaches also demonstrated statistical superiority. Futakuchi et al. (Futakuchi et al., 2021) developed a Deep Q-network (DQN)-profit sharing based battery scheduling model, reducing battery capacity requirements by up to 7.75 h. Bui et al. (Bui et al., 2020) applied multi-Agent DRL to maximize wind farm output under wake interactions, achieving 1.99–4.11 % improvement over Maximum Power Point Tracking (MPPT). While these studies did not isolate SCADA noise, their adaptive training indicates strong robustness to fluctuating signals. In a security-focused context, Liu et al. (Liu et al., 2020) incorporated DRL with Common Vulnerability Scoring System (CVSS) metrics for resilient behavior modeling, highlighting DRL’s potential in partially observable, noisy SCADA systems.

Saini et al. (Saini et al., 2020) compared ELM, NARX, and LSTM for wind speed forecasting and showed LSTM’s superior robustness, achieving the lowest MAPE (0.2298) and RMSE (1.0594). Jiménez et al. (Jiménez et al., 2020) further validated the NARX- Quadratic Discriminant Analysis (QDA)-ensemble pipeline with wavelet-filtered SCADA signals, showing over 99 % accuracy and statistical significance via Friedman and Bonferroni-Dunn tests. Mohandes and Rehman (Mohandes and Rehman, 2018) employed DNNs to extrapolate wind speeds to hub height, achieving MAPE = 9.65 % and $R^2 = 0.94$ under incomplete SCADA scenarios. Lin and Liu (Lin and Liu, 2020) employed a five-layer DNN trained on high-frequency offshore SCADA data, achieving RMSE = 517.33 and $R^2 = 0.91$, while identifying feature redundancies via sensitivity analysis. Mishra et al. (Mishra et al., 2020) applied multivariate time-series modeling with Fast Fourier Transform (FFT)- and wavelet-transformed data, where attention models achieved the lowest MSE and RMSE values across 1–12-month horizons, confirming the superiority of hybrid preprocessing and transformer-based learning. Dhiman and Deb (Dhiman and Deb, 2021) found CNNs most effective for long-term ramp prediction in noisy conditions, though twin SVR performed best with limited training data.

Salah et al. (Salah et al., 2022) compared regression and ML models and found RF and LSTM outperforming others in East Jerusalem’s unique climate, with RF achieving MAE = 0.894 m/s. However, the study lacks hybridization and ensemble strategies. Khan et al. (Khan et al., 2020) proposed a hybrid MLP–SVR–CatBoost model for energy forecasting in Jeju Island, achieving MAPE = 4.29 %, clearly outperforming individual learners. This fusion approach proved statistically reliable across all time windows and energy sources.

Across all studies, hybrid models combining signal decomposition like EWT, VMD and EEMD and DL architectures like LSTM, CNN and deep Boltzmann machine consistently outperformed traditional and shallow models in noisy SCADA environments. Decomposition helps mitigate non-stationarity and noise, while temporal models capture long-range dependencies. Additionally, optimization techniques (PSO, HHO, GA, WOA) and ensemble strategies such as DQN, Multi-Agent DRL, and hybrid CNN–SVM frameworks introduced adaptability and improved generalization, especially when backed by robust validation metrics like MAPE, RMSE, AUC, and Friedman tests. CNNs demonstrated exceptional performance in both predictive maintenance and fault diagnostics, with AUCs exceeding 0.99 in some cases (Bach-Andersen et al., 2018), indicating industrial-scale applicability.

However, several gaps remain. While many studies employ extensive statistical validation, only a subset integrates cross-validation, formal hypothesis testing, or multi-site generalization, which are necessary for real-world reliability. Ensemble averaging (Piotrowski et al., 2022),

PCA-based dimensionality reduction (Khan et al., 2019), and transformer models (Mishra et al., 2020) show emerging promise but are not yet widely compared across datasets. Furthermore, while DL and RL models show superior performance, explainability and interpretability are often overlooked—limiting their transparency in operational decision-making. Future research should prioritize benchmarking under standardized noisy SCADA datasets, integration of uncertainty quantification, and explainable AI mechanisms to close the gap between model performance and deployability in wind energy systems.

4.4. RQ4: What are the potential benefits and drawbacks of using machine intelligence approaches in wind energy utilization compared to traditional approaches?

This section evaluates the benefits and drawbacks of machine intelligence methods, especially supervised and optimization-based techniques, in comparison to traditional models.

ML techniques have demonstrated advantages over traditional statistical and physical models in wind energy utilization, particularly in forecasting, control, and optimization. Ensemble learning, DNNs, and hybrid models have enhanced forecasting accuracy and robustness. For instance, Crespo-Vazquez et al. (Crespo-Vazquez et al., 2018) developed a decision-making framework for managing energy price uncertainty, while Jalali et al. (Jalali et al., 2021) applied an ANFIS with PSO and GA to improve prediction accuracy.

Hybrid approaches such as those by Khan et al. (Khan et al., 2019), Yi et al. (Yi et al., 2019), and Santhosh et al. (Santhosh et al., 2019) used methods like PCA, LSTM, and ELM to enhance generalizability and reduce overfitting. DL models have also supported offshore wind forecasting and extrapolation at higher altitudes (Mohandes and Rehman, 2018; Lin and Liu, 2020). Qu et al. (Qu et al., 2018) employed GAN and CNN for data imputation, improving data completeness and forecasting reliability. RL and DRL frameworks have shown success in control strategies, such as adaptive dispatch (Jeong and Kim, 2021) and hybrid quantification under uncertainty (Lin and Liu, 2020).

However, limitations persist. Many ML models are data-intensive and computationally demanding, limiting scalability in real-time or resource-constrained environments. Deep models can also suffer from interpretability issues, complicating transparency and trust. Additionally, inconsistent data quality, overfitting, and sensitivity to hyperparameters reduce the generalizability of models across turbine types and locations.

Despite these drawbacks, the flexibility and adaptability of ML models represent a major leap forward. Their ability to integrate real-time data, optimize performance, and support condition monitoring makes them critical for modern wind energy systems.

Key Advantages

- ML models, especially ensemble and hybrid frameworks, outperform traditional approaches in handling nonlinearities and spatiotemporal variability (Crespo-Vazquez et al., 2018; Khosravi et al., 2018; Zhu et al., 2023).
- RL and DRL models adapt to dynamic conditions, enhancing decision-making in control and scheduling tasks (Jeong and Kim, 2021; Liu et al., 2022).
- DL models handle high-dimensional SCADA data, enabling robust fault detection and predictive maintenance across large wind farms (Lin and Liu, 2020; Qu et al., 2018).
- Forecast accuracy improves grid stability, energy dispatch, and scheduling, especially under intermittent resource conditions (Sun et al., 2019; Grève et al., 2020).
- ML facilitates smart grid compatibility and supports distributed resource coordination (Grève et al., 2020).

Key Limitations

- Forecast accuracy relies heavily on the volume and quality of training data. Poor sensor readings or missing data can degrade performance (Zhao et al., 2018; Qu et al., 2018).
- Deep models often act as black boxes, making it difficult to trace predictions back to physical phenomena (Liu et al., 2018; Yeh et al., 2019).
- Resource-intensive models like CNN and GAN can be impractical for small operators or low-latency systems (Mohandes and Rehman, 2018; Chen et al., 2018).
- ML models trained on specific sites may not generalize well to others due to regional environmental variability (Díaz et al., 2018; Sun and Sun, 2018).
- Many models remain at the simulation stage without validation in large-scale commercial deployments (Sáenz Díaz, 2022; Zhang et al., 2019).

Table 4 and Table 5 discuss the advantages and limitations of the latest study mainly related to power generation, prediction, fault diagnosis and design. Table 6 shows the categorization of AI-based approaches for wind energy systems – methods, domains, and applications.

4.5. Methodological classification of reviewed techniques

Machine intelligence techniques adopted in wind energy systems span a wide spectrum of modeling paradigms, ranging from traditional ML to DL, hybrid models, and reinforcement learning. This section categorizes the reviewed approaches based on their methodological structure and primary application domain, serving as a framework to understand their respective strengths and use cases.

Shallow machine learning models

Shallow models such as SVM, DT, KNN, NB, and RF are commonly used for tasks including wind power forecasting, anomaly detection, and classification (Crespo-Vazquez et al., 2018; Wang et al., 2023; Díaz et al., 2018; Liu et al., 2023; Bin Abu Sofian et al., 2024; Oyucu and Aksöz, 2024; Mollick et al., 2024). These techniques are preferred for their simplicity, low computational requirements, and moderate interpretability, making them suitable for rapid prototyping and resource-constrained environments.

Deep learning models

Deep learning methods including CNN, LSTM networks, GRU, and DBN—have demonstrated strong performance in tasks involving temporal pattern recognition, spatial feature extraction, and image-based fault analysis (Liu et al., 2020; Mohandes and Rehman, 2018; Jiang et al., 2022; Chen et al., 2018; Penchalaiah and Ramya, 2023; Ozbek et al., 2022; Biswas and Sinha, 2021; Liu et al., 2023; Kumar and Kumar, 2024; Anushalini and Sri Revathi, 2024; Buestán-Andrade et al., 2024; Oyucu and Aksöz, 2024; Sahin and Ozbay Karakus, 2024; Tang et al., 2024; Behara and Saha, 2024). These models are particularly effective for time-series forecasting and multivariate SCADA data analysis.

Hybrid architectures

Hybrid models integrate complementary machine learning paradigms to enhance performance and generalizability. Examples include PCA–DL for dimensionality reduction and prediction (Khan et al., 2019), EWT–LSTM for noise-resilient time series modeling (Liu et al., 2020), VMD–LSTM for decomposed sequence forecasting (Liu et al., 2018), and CNN–GRU–Attention for long-term dependency modeling (Hossain et al., 2021). Optimization-enhanced hybrids, such as ANFIS–PSO and Transformer–FFT–Wavelet (Khosravi et al., 2018; Mishra et al., 2020), further improve adaptive capability.

Ensemble models

Ensemble methods such as RF, Gradient Boosting, XGBoost, LightGBM, and AdaBoost aggregate the predictions of multiple base learners to improve accuracy and robustness (Oyucu and Aksöz, 2024; Mollick et al., 2024; Zhao and Zhou, 2024). These models are widely used in classification and regression tasks involving SCADA data and blade condition monitoring.

Table 4

Advantages and limitations of the approaches used mainly in wind power generation and energy optimization.

S. No	Source	Method used	Area	Advantages	Limitations
1	(Penchalaiah and Ramya, 2023)	Enhanced Garra Rufa Fish Optimization and Artificial Transgender Longicorn Algorithm for control scheme optimization and adjustment of electrical system stabilizer parameters	Wind Energy Systems	Improves stability via parameter tuning	Needs real-world validation
2	(Chen et al., 2019)	DNN with triplet loss	WT fault detection (blade icing accretion) using imbalanced SCADA data	Effectively handles imbalanced data using triplet loss and learns deep feature embeddings preserving both within-class and between-class information.	Only considers between-class imbalance, not intra-class variation
3	(Hao et al., 2023)	DL combined model based on mixed frequency modelling	Wind energy, renewable energy, urban energy systems	Improves forecast accuracy using mixed data	High computational demands
4	(Chen et al., 2018)	DL and IoT	Wind power generation	Combines physical and data-driven models	The proposed approach may require significant amounts of data to train the DL model, which may be difficult to obtain in some cases
5	(Shetty et al., 2021)	A multi-step hybrid online wind speed forecasting model that combines online sequential ELM, optimized VMD and cuckoo search optimization algorithm	Wind power generation	Better nonlinear pattern capture	Complex implementation
6	(Ozbek et al., 2022)	Four different ML algorithms are proposed, including ANFIS with Fuzzy c-means, ANFIS with Grid Partition, ANFIS with Subtractive Clustering, and LSTM NN	Wind energy forecasting	Useful for the efficient planning and operation of wind energy generation.	The study is limited to only three measurement stations in Turkey, and the results may not be generalized to other locations with different weather patterns
7	(Pujari et al., 2023)	LSTM networks, Nonlinear AR models and Wavelet Neural Networks and an evolutionary multi-objective strategy	Wind energy forecasting	Low-cost and generic DNN design	Needs broader scenario testing
8	(Yang et al., 2020)	Penalized ELM	Ultra-short-term forecasting	Handles outliers, efficient ANN structure	Limited to short-term context
9	(Liu et al., 2022)	A combined model on data preprocessing, linear prediction model, ANN, DL and multi-objective optimization	Wind energy and wind speed prediction	Stable and precise forecasts	Needs validation on broader datasets
10	(Guo et al., 2022)	Hybrid prediction method combining EWT, multi-objective modified seagull optimization algorithm and multi-kernel ELM	Wind power generation, wind speed prediction	More accurate and stable predictions	The study does not address the potential computational cost or its implementation in real-time wind power generation systems
11	(Deepa and Banerjee, 2022)	The nonlinear RBFNN model and wavelet neural network model are developed and modified to improve wind speed forecasting accuracy	Wind power generation, specifically wind speed forecasting	Enhanced forecast precision	Needs field testing
12	(Biswas and Sinha, 2021)	DL architectures based on the LSTM model and bidirectional LSTM model for wind speed time series	Wind speed forecasting	Captures irregular patterns well	Ignores some meteorological factors
13	(Lv and Wang, 2023)	Filter-wrapper non-dominated sorting differential evolution algorithm incorporating K-medoid clustering to select key meteorological factors, hybrid DL model with SSA and convolutional LSTM network	Wind speed forecasting with consideration of multiple meteorological factors	Accurate with relevant meteorological features	Regional transferability unclear
14	(Devi et al., 2020)	Improved LSTM with cuckoo search optimization algorithm	Wind power forecasting using ensemble empirical mode decomposition	Operational tool for energy supply	Sensitive to weather variation
15	(Singh and Rizwan, 2022)	SVR and gradient-boosting regression trees	Wind power production forecasting	Improves short-term accuracy	Needs cross-data validation
16	(Kim et al., 2022)	ML models	Power production forecasting	Accurate with minimal inputs	Omits key environmental factors
17	(Jiang et al., 2022)	A hybrid framework combining CNN, Attention-based LSTM network, and AR model	Wind power generation forecast of multiple RES	Captures local and global patterns	Limited generalization
18	(Li and Zou, 2022)	ELM + PSO	Short-term forecasting	Lower error across sub-signals	High computational cost
19	(Yang et al., 2019)	The study proposes a data-driven approach based on supervised learning.	Wind energy and weather forecasting	Captures spatial-temporal variations	Relies on low-height wind data
20	(Jalali et al., 2021)	The paper proposes a novel framework for optimizing the hyperparameters and architecture of DNNs	Wind speed forecasting for wind power utilization	No expert tuning needed	High training resource cost
21	(Baile and Muzy, 2023)	Model comparison with spatiotemporal data	Short-term prediction	Nonlinear models outperform LR	Lacks environmental variable inputs

(continued on next page)

Table 4 (continued)

S. No	Source	Method used	Area	Advantages	Limitations
22	(Liu et al., 2022)	ML data reconstruction model using the RF method	Wind speed field reconstruction	Better than interpolation methods	Sensitive to training data quality
23	(Chen et al., 2018)	DL + preprocessing + tuning	Predictive maintenance	Handles big data challenges	Scalability not discussed
24	(Ozbek et al., 2022)	ML-derived surrogate model and ANN-based optimizer	Wind engineering and computational modelling for pressure field and sensor placement optimization	Fast and low-cost pressure prediction	Accuracy depends on training set
25	(Al-Janabi et al., 2020)	A model called Multi-Objectives Renewable Energy Generation to generate electrical energy from wind	Energy optimization	Improves energy generation efficiency	Needs real-world scalability test
27	(Ashwin Renganathan et al., 2022)	WT LiDAR measurements are used to construct predictive surrogate models using deep autoencoders and a DNN to learn the mapping between the parameter space and the wake flow fields. GP was applied to learn the parameter-space-latent-space mapping	Wind farm design optimization using ML	Cheaper than physics-based simulations	Limited terrain/wind condition fit
28	(Behzadi et al., 2023)	Rule-based control	Energy management in intelligent building systems	Cuts peak load and emissions	Limited generalizability

Signal decomposition with machine learning

Signal decomposition techniques such as EWT, VMD, SSA, and OAIMF are used to preprocess wind speed and SCADA signals, enhancing the performance of downstream ML models (Liu et al., 2020; Liu et al., 2018; Mortezaazadeh et al., 2022; Behara and Saha, 2024). These methods extract multi-scale temporal features and improve learning under non-stationary conditions.

Optimization-based tuning

Optimization algorithms like PSO, GA, HHO, GWO, WOA, and TLBO are employed to fine-tune ML/DL models. These strategies facilitate feature selection, hyperparameter tuning, and weight initialization, thereby improving convergence and model performance (Khosravi et al., 2018; Hossain et al., 2021; Jiang et al., 2022; Liu et al., 2022; Guo et al., 2022; Devi et al., 2020; Ashwin Renganathan et al., 2022; Sahin and Ozbay Karakus, 2024).

Reinforcement learning

RL, especially Deep RL techniques such as DQN, DDQN, DDPG, and multi-agent frameworks, are increasingly applied to adaptive control tasks in wind energy systems. These include yaw angle regulation, battery scheduling, and wake flow management (Crespo-Vazquez et al., 2018; Song et al., 2024; Li et al., 2024; Liu et al., 2023; Wang et al., 2024; Chengqing et al., 2023; Kumar and Kumar, 2024; Liu et al., 2023; Li et al., 2024; Tang et al., 2024; Wang et al., 2025). RL models learn optimal actions through trial-and-error, making them well-suited for dynamic environments.

Physics-informed machine learning

Physics-informed ML represents a promising integration of CFD, RANS models, and wake flow simulations with data-driven learning frameworks (Richmond et al., 2020; Mortezaazadeh et al., 2022; Ti et al., 2020; Ashwin Renganathan et al., 2022; Moss et al., 2024). These approaches combine the empirical accuracy and interpretability of physics-based simulations with the flexibility and generalization capabilities of ML. These methods are particularly suited for offshore wind farm modeling, wake effect prediction, and rotor-level flow estimation, where capturing complex flow interactions and maintaining physical consistency is essential.

A notable advancement within this domain is the development of physics-informed neural networks, which incorporate governing physical laws directly into the loss function of DL models. These networks enable learning from sparse data while preserving physical constraints, making them attractive for flow field reconstruction, load estimation on blades, and real-time simulation surrogates.

Compared to conventional DL models like CNNs or LSTMs, physics-informed frameworks offer superior interpretability, less susceptibility

to overfitting, and the ability to generalize under data-sparse or unseen physical conditions. However, they often suffer from high computational costs, complex implementation, and slower training convergence, particularly in large-scale 3D CFD settings. Additionally, their application in real-time control and optimization remains underexplored and poses an important direction for future research in WT automation and digital twin development.

4.6. Performance benchmarking and critical comparison of machine intelligence techniques in wind energy systems

Machine intelligence techniques used in wind energy systems vary substantially in their ability to predict, generalize, adapt to noise, and operate under real-time constraints. This section consolidates performance evaluation metrics and presents a critical comparative assessment of various ML, DL, hybrid, and RL approaches.

4.6.1. Performance benchmarking metrics and overview

The integration of machine intelligence techniques into wind energy systems has enabled advancements across forecasting, control, fault detection, and performance optimization. However, due to the diversity of application areas, rigorous performance benchmarking requires that appropriate metrics be aligned with specific tasks. Forecasting tasks are typically evaluated using regression-based metrics such as MAPE, RMSE, MAE, and the R^2 , which quantify predictive accuracy over continuous variables like wind speed or power output. Classification tasks, particularly those targeting fault detection, damage assessment, or maintenance forecasting, are generally evaluated using accuracy, precision, recall, F1-score, and the AUC. For optimization and control tasks, metrics often include power improvement percentages, dispatch efficiency, and energy savings.

4.6.2. Forecasting and prediction tasks

Accurate wind power and wind speed forecasting are essential for efficient turbine scheduling and grid stability. Over the last decade, numerous machine intelligence models have been proposed, with hybrid and DL techniques consistently outperforming classical statistical approaches. Among the best performers, Santhosh et al. (Santhosh et al., 2019) achieved a remarkably low MAPE of 0.9941 % and RMSE of 0.1238 using an EEMD-deep Boltzmann machine framework. Liu et al. (Liu et al., 2018) reported a MAPE of 6.49 % and RMSE of 0.4038 using a VMD-SSA-LSTM-ELM hybrid, highlighting the advantage of combining decomposition and sequential modeling. Transformer-based models also yielded strong results; Mishra et al. (Mishra et al., 2020) demonstrated

Table 5

Advantages and limitations of the approaches used mainly in prediction, fault diagnosis and design.

S. No	Source	Method used	Area	Advantages	Limitations
1	(Behzadi and Sadrizadeh, 2023)	Hybrid solar-wind with hydrogen storage (TRNSYS simulation tool, rule-based control)	Residential energy system (Sweden)	Effective hybridization, increased grid stability, lower peak load, energy savings, carbon dioxide reduction, and overall cost reduction	Maintenance of fuel cell voltage and collector length at their lowest domain, the ineffective parameter of electrode area, and solar energy's limited effectiveness in storing hydrogen over winter
2	(Eslamirad et al., 2023)	ML with a data-driven approach based on the classification	Predicting outdoor thermal comfort in cities	Data-driven and generative approach to produce reliable data on outdoor comfort using ML	Limited to the specific context of Tallinn, Estonia, and the opaque ML model may limit interpretability
3	(Kapp et al., 2023)	Supervised ML model (linear regressor)	Industrial energy consumption prediction	Uses multiple features to predict energy consumption and can be applied to different manufacturing plants	Relies on available data from industrial energy audits and may not capture all factors contributing to energy consumption
4	(Crespo-Vazquez et al., 2018)	Federated DRL	Wind power prediction for power grid dispatching plans in the field of power system management	Data privacy, improved accuracy, communication efficiency, and confirmed robustness	Demands high computation, limited to short-term
5	(Liu et al., 2020)	Wind speed combined probability prediction system integrating data denoising and quantile-based DL models	Wind speed prediction for grid connection safety and wind farm management	Improves prediction accuracy, builds appropriate prediction intervals, and measures and minimizes uncertainty	Different other scenarios can be considered
6	(Wang et al., 2023)	Heuristic Fuzzy Logic-Based Controller using ML models (ANN and SVM) for frequency control in wind energy systems	Frequency control and load forecasting in wind-based hybrid power systems	Effective frequency control support, modelling of complex system behaviour, improved load forecasting, and reduced prediction errors	The effectiveness of the method may depend on the quality and availability of data
7	(Harris et al., 2023)	ML methods include Reconstructability Analysis, Bayesian Networks, SVR and ANN	Load imbalance prediction	Cost-effective, reduces ramping reserves	Capture efficiency can be optimized
8	(Balakumar et al., 2023)	RF + DT + pricing model	Smart building energy management and demand side management for energy flexibility	Enhances demand response, user benefits	May not generalize to all cases
9	(Liu et al., 2023)	SVM	Fault diagnosis in mine ventilation systems using ML	Rapid diagnosis and mitigation of windage alteration faults (WAFs) in mine ventilation systems	The generalization performance of the model is significantly affected by sample attributes
10	(Wang et al., 2023)	Multi-objective DRL models (MO-DDPG and MO-DSPG)	WT blade design optimization	Improved aerodynamic performance, accurate solutions on Pareto fronts, and improved computational efficiency	The overestimation of the Q-value can be a limitation
11	(Behzadi and Sadrizadeh, 2023)	Generalized multiscale Poincare plots captures behaviour changes in vibration signals through scales. Support vector data description is then used to learn the acceptance region defined for anomaly detection under normal conditions	WT fault detection using vibration signals	Early anomaly alerts, effective dynamic tracking	Needs representative normal samples
12	(Penchalaiah and Ramya, 2023)	Semi-supervised DL	WT systems fault recognition	The proposed method can effectively identify new class faults in WT systems that traditional fault recognition algorithms cannot identify	The computational complexity and time required for training the model
13	(Jiang et al., 2020)	Feature extraction + PCA + optimized DT	WT bearing fault diagnosis	High accuracy in detecting bearing issues	Feature extraction may reduce data fidelity

that integrating FFT and wavelet-transformed inputs with a transformer architecture led to the lowest MSE and RMSE across 1–12-month forecasting intervals. Notably, Hossain et al. (Hossain et al., 2021) applied HHO to a CNN–GRU model, reducing MAPE by 37.8 % and 24.1 % for 5-minute and 10-minute forecasts, respectively.

Edge-deployable and hardware-efficient solutions have also gained momentum. Buestán-Andrade (Buestán-Andrade et al., 2024) evaluated CNN, GRU, fully connected, and transformer models on SCADA data for real-time forecasting on a Raspberry Pi. CNN emerged as the most balanced model, achieving RMSE = 0.0596, MAE = 0.0330, and weighted MAPE = 0.0852, while consuming only 332.5 μ Wh/prediction. GRU had slightly better accuracy but at a significantly higher energy cost. In the context of wind farm control, Liew et al. (Liew et al., 2024) developed a model-free RL strategy using RLS to optimize turbine yaw angles, leading to power gains of up to 11.6 ± 3.0 % in partial-wake

conditions. For blade health monitoring, Altice et al. (Altice et al., 2024) used deep CNN(DCNN) and transfer learning on the CAI-SWTB dataset, with TL-Xception achieving 99.92 % accuracy on small turbine blades and 100 % generalization accuracy on large-scale blades. Hybrid and domain-specific models further enhance prediction robustness. Chen et al. (Chen et al., 2019) applied a DL-based diagnostic model incorporating autoencoders and grey dynamic prediction, enabling fault detection up to 40 min in advance with stable recovery under 30 % torque deviation scenarios. Manshadi et al. (Dehghan Manshadi et al., 2022) used LSTM and other ML models to model offshore hybrid wind–wave energy systems, achieving up to 0.97 accuracy with RMSE = 16.4. Additionally, Qu et al. (Qu et al., 2018) addressed missing SCADA data issues using conditional GANs and CNN-based architectures, outperforming traditional imputation techniques by effectively reconstructing incomplete WT datasets. The proposed method has the lowest

Table 6

Categorization of machine intelligence approaches for wind energy systems – methods, domains, and applications from 2023 to 2024.

S. No	Source	Methods	Category	Dataset / Domain	Results	Advantages	Limitations	Implications and Practical Implementation Challenges
1	(Song et al., 2024)	Adaptive GA + Deep Deterministic Policy Gradient (AGA-DDPG)	DRL-Based Multi-Agent Optimization	Offshore Wind Farms, Simulated Multi-access Edge Computing (MEC) Environment with Unmanned Aerial Vehicles (UAVs)	Reduced overall system overhead compared to Lagrangian Control (LC), Reactive Control (RC), Twin Delayed Deep Deterministic Policy Gradient (TD3), and Deep Deterministic Policy Gradient (DDPG) by 61.8 %, 55 %, 21 %, and 33 %, respectively	Improved convergence speed and offloading efficiency, adaptive task scheduling, better resource utilization	Performance may vary in the real world due to idealized assumptions	Enhances real-time monitoring and predictive maintenance in offshore wind farms and supports carbon neutrality goals. High computational complexity due to Adaptive Genetic Algorithm (AGA); real-world deployment requires dynamic adaptation, field validation, and real datasets
2	(Li et al., 2024)	DL-RL-Model Predictive Control (MPC) (RBFN + DQN + MPC); hybrid data-driven control with online power strategy	DRL-enhanced Predictive Control	IEA 15-MW offshore WT (Task 37), simulated using OpenFAST and TF-Agents	Achieved constant power tracking with up to 92 % reduction in power fluctuation; robust maximum power capture comparable to MPPT; enhanced two-degree-of-freedom (2-DOF) optimization accuracy (>98 %)	Enables flexible turbine operation (not limited to MPPT) and intelligent 2-DOF control, reduces power oscillation, is robust to varying power targets, and has an enhanced output quality	Computationally intensive; RL model requires large training iterations (~30,000); not real-time ready for sudden wind changes; relies on accurate wind forecasting	Intelligent, self-optimizing turbines that adapt to grid demands and contribute to smart grid stability. High-frequency update of RL-based Optimal Power Strategy (OPS) is challenging; wind speed forecasting is needed; DL-RL-MPC must be tuned per turbine; validation in real-world turbine is required. Demonstrates scalable intelligent Energy Management System (EMS) supports resilient, economic grid operations with DRL and uncertainty-aware control. Accurate real-time input data required (price, wind, SOC); deployment demands model tuning and forecasting integration
3	(Liu et al., 2023)	DRL with wind power prediction intervals (LSTM-Lower Upper Bound Estimation (LUBE) + DQN)	DRL-Based Energy Storage Management	Real wind farm in Turkey (2018 data, 8016 hourly samples)	Achieved 0 % wrong Energy Storage System (ESS) decisions; improved net economic benefit while reducing wear cost; maintained State of Charge (SOC) above emergency reserve with high reliability	Quantifies wind power uncertainty; data-driven ESS scheduling; improves reliability and economics; robust multi-step prediction performance	Training DRL and LSTM-LUBE models is computationally intensive; model generalizability to different grid setups is not validated	Introduces a generalizable, intelligent control strategy for next-gen flexible WECs; enables resilient, high-efficiency wave energy extraction under uncertainty. Wave prediction systems and sensors needed; tuning needed per WEC type; inference time and DNN stability crucial in real deployments
4	(Wang et al., 2024)	DDQN-based non-causal control with wave prediction for point absorber Wave Energy Converters (WECs)	DRL + Non-causal Control	Real wave data from Cornwall, UK; also tested on synthetic waves	3–5 step prediction improved energy output $3.6 \times$ compared to traditional proportional-derivative controller; robust against prediction noise ($\text{SNR} \geq 10$ dB) and model uncertainties ($K \leq 0.1$)	Adaptive gain tuning; handles highly nonlinear or unmodelled WECs (e.g., DEGs, origami WECs); non-causal prediction-based control enhances energy extraction	Requires long training (up to 12 hrs); sensitivity to prediction model quality; effectiveness depends on prediction horizon and wave data quality	

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Table 6 (continued)

S. No	Source	Methods	Category	Dataset / Domain	Results	Advantages	Limitations	Implications and Practical Implementation Challenges
5	(Chengqing et al., 2023)	Graph Attention Network- Gated Recurrent Unit – Temporal Convolutional Network (TCN)-Deep Deterministic Policy Gradient (GAT-GRU-TCN-DDPG)	Ensemble DL with RL	Wind power, weather, turbine state data (real-world data from wind farms in Xinjiang, China)	Outperformed 26 benchmark models: improved performance by > 10 % over traditional methods	Captures spatiotemporal dependencies, integrates multi-source data, high generalization and robustness, dynamic ensemble learning	Requires predefined graph structure, long training time, and not fully optimized feature selection	Useful for accurate wind power forecasting, wind farm scheduling, and energy planning. Computationally intensive; sensitive to quality of spatial and temporal data; real-time deployment challenges
6	(Kumar and Kumar, 2024)	Hybrid Autoregressive Integrated Moving Average (ARIMA)-DRL (HARIMA-DRL)	Resilience Enhancement in Microgrids	ACTIVSg2000 test system, synthetic disaster data (5000, 8000, 10,000 samples)	HARIMA-DRL improves resilience by 3.07 %, 7.78 %, and 11.51 % over GRM and CNN-LSTM for 5000, 8000, and 10,000 datasets, respectively; RMSE as low as 0.0332	Smooth nonlinear prediction, flexible kernel settings, accurate forecasting, adaptive control, robust to disruptions	Depends on training data quality, complexity of implementation, and real-time adaptability	Effective disaster recovery planning, improved reliability of decentralized energy systems, potential to extend to other critical infrastructures. Real-time retraining, disaster simulation fidelity, integration with real microgrid control systems
8	(Liu et al., 2023)	CorrDQN-FS (Correlation + DQN-based DRL Feature Selection)	Feature Selection for Energy Prediction	Real-world energy data from a commercial office building in Shanghai (Hourly data, 2015–2016, 17,520 samples)	CorrDQN-FS, combined with DF-DDPG, achieved the best prediction accuracy with only 4 features selected.	Captures both linear and nonlinear dependencies; significantly improves prediction accuracy; reduces dimensionality.	Needs well-tuned hyperparameters; computational complexity of DRL	Enhances intelligent energy management; applicable to building anomaly detection and sustainability goals. Requires integration of DRL agent with predictive models; demands high-quality data preprocessing
9	(Li et al., 2024)	Meta Multi-Agent Reinforcement Learning (Meta-MARL) combining Meta-RL & MARL to solve Multi-objective Two-Stage Robust Optimization (MOTSRO)	DRL, Multi-agent Systems, Optimization	IEEE 39-bus system; Real-world wind/PV/load & price data from Shanxi, China (2023)	Outperforms in Pareto front convergence, diversity, and uniformity; improves scheduling robustness and reduces computation.	The end-to-end model handles uncertainty and multiple objectives. Robust first-stage strategy. Fast convergence using Meta-RL Modular, scalable design.	Dual transformation is needed for second-stage limit generalization. It cannot solve TSRO directly without convexity assumptions.	Supports robust, long-term energy dispatch decisions in uncertain multi-energy systems and the potential for expansion to larger grids with distributed/hierarchical RL.
10	(Pradeep et al., 2024)	Hybrid Forecasting System: K++ Clustering + Teaching-Learning Based Optimization (TLBO) + Deep SARSA	Hybrid DRL, Feature Selection, Forecasting	Real-world onshore wind farm data; Simulated with high-resolution weather research & forecasting model	Improved prediction accuracy for hybrid renewable power generation; enhanced forecasting of wind energy	Integrates meteorological modeling with AI-based forecasting. Handles faulty data via clustering. Optimal feature selection using TLBO	– Lacks detailed quantitative evaluation metrics. – May require high computational resources for simulations	– Real-time data uncertainty – Scalability to national-scale grids- Reliable real-world reward function modeling Supports reliable power prediction in hybrid systems, with potential for improved grid planning and dispatch strategies. Implementation of high-resolution models and integration with real-time hybrid systems, although it is needed robust online learning

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Table 6 (continued)

S. No	Source	Methods	Category	Dataset / Domain	Results	Advantages	Limitations	Implications and Practical Implementation Challenges
11	(Sofian and A. D.A., Lim, H. R., Siti Halimatul Munawaroh, H., Ma, Z., Chew, K.W., Show, P.L., 2024)	Review of ML applications: RF, ANN, SVR, DRL, SVM, MLP, ACO, WT, etc.	Review – ML in wind/solar energy systems and forecasting	Multiple real-world and simulation-based datasets across different studies (e.g., MERRA-2, sensor data, historical wind data from Turkey, Tunisia, etc.)	ML significantly improves forecasting accuracy, enables optimization of renewable energy systems, supports hybrid systems, and enhances energy storage integration	Extensive compilation of ML approaches for energy prediction and management; Emphasizes integration with smart grids, energy storage, and EVs.	High complexity in real-world deployment; Intermittency of sources remains an issue	Offers strategic direction for AI-driven sustainable energy adoption; Highlights the need for future work on storage, AI optimization, smart infrastructure. Real-world deployment of ML in hybrid systems; data quality, integration with physical systems, and environmental and social factors
12	(Anushalini and Sri Revathi, 2024)	Baseline, Linear, Dense, Multi-Dense, CNN, LSTM, Residual LSTM	Time-series prediction of wind power using ML/DL	Historical wind data: inputs, e.g., temperature, pressure; output – hourly wind power generation	LSTM and Residual LSTM models outperformed others with MAE of 0.0987 and 0.0958, respectively	In a comprehensive comparison of ML and DL models, Residual LSTM yielded the best accuracy.	Model performance was only evaluated using MAE; it may benefit from broader metrics like RMSE or R^2 .	Demonstrates potential of DL (especially LSTM variants) in improving wind power forecasting accuracy. Model generalizability to other regions or systems requires quality historical data for training.
13	(Buestán-Andrade et al., 2024)	CNN, fuzzy clustering, GRU, Transformer	ML/DL-based time-series forecasting	SCADA data from a WT in Turkey (10-min intervals; features: wind speed, direction, power curve, active power)	CNN achieved the best trade-off between accuracy and efficiency on a low-cost device (Raspberry Pi 3); GRU had the lowest MAE but higher energy usage	Demonstrated feasibility of accurate wind forecasting on Raspberry Pi; evaluated energy usage and prediction speed.	GRU had high energy use and lower real-time performance; Transformer had higher error on some metrics	Enables decentralized wind energy management and democratizes access to forecasting tech, especially in off-grid settings. Computational constraints on low-cost hardware; need for efficient model optimization.
14	(Oyucu and Aksöz, 2024)	LR, DT, RF, GBM, XGBoost, LightGBM, CatBoost with MLOps	ML, MLOps Integration	SCADA data from a WT in Türkiye	LightGBM (after tuning): RMSE reduced to 190.34 kW; Latency: 9 ms via Docker deployment	Integrated MLOps pipeline for end-to-end automation; improved deployment speed, scalability (via Docker Swarm), and accuracy	Initial LightGBM tuning had limited gain; it requires domain-specific feature engineering; MLOps needs DevOps expertise.	Enhances the accuracy and speed of wind energy forecasting and enables operational deployment and scaling of ML models. Automating Continuous Integration and Continuous Deployment (CI/CD); managing model lifecycle, hyperparameter tuning is required for best performance.
15	(Moss et al., 2024)	RF pseudo-2D Reynolds-Averaged Navier–Stokes (RANS) model, RF-based surrogate modeling, DeepHyper optimization	ML and Physics-Based Modeling	SCADA data from King Plains wind farm (AWAKEN site), Oklahoma, USA	RF model NMAE: 5.3 % (farm), 15.9 % (turbine); RANS surrogate NMAE: 11.6 % (farm), 22.1 % (turbine)	RF models are more accurate than RANS surrogate in both farm and turbine predictions; Lower computational cost for RF; Can capture intra-farm wake variability better	Both models struggle to accurately predict power loss due to farm-to-farm interactions; RANS underestimates, RF overestimates variability	Highlights strengths of ML in handling wake-induced variability; Encourages hybrid RANS + ML modeling for complex inflows. Defining accurate freestream input conditions, model tuning under varying

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Table 6 (continued)

S. No	Source	Methods	Category	Dataset / Domain	Results	Advantages	Limitations	Implications and Practical Implementation Challenges
16	(Liew et al., 2024)	Model-Free Closed-loop Control Using RL (Policy Gradient) + Recursive Least Squares (RLS)	RL-based Wind Farm Control	Simulated 4-turbine wind farm using HAWC2Farm (DTU 10 MW turbine model)	Power gains: 11.6 ± 3.0 % (partial wake), 1.4 ± 0.5 % (full wake) at $TI = 7.5$ %	Model-free approach adapts to dynamic conditions; RLS is computationally efficient; Improved power capture under wake interactions	Long learning time (up to days); performance degrades with high turbulence; cannot recall past states due to RLS memory decay	atmospheric stability, and generalization to unseen farm layouts Demonstrates viability of data-driven control for wake steering; potential for hybrid model-informed RL to improve convergence and robustness. Real-time tuning of many control parameters, achieving convergence before atmospheric conditions shift.
17	(Mollick et al., 2024)	DT, RF, XGBoost, KNN, Extra Trees, CatBoost, SVR, Lasso, Ridge, etc.	Wind Speed Prediction, Site Selection	Wind speed data from 2000 to 2020 for four coastal Indian sites (e.g., Digha, Haldia)	Ensemble models (e.g., XGBoost, RF, Extra Trees) outperformed others	Evaluates 12 ML models for robust site selection; identifies best-performing models for specific regions	Lack of uncertainty quantification, limited geographic generalizability, no hybrid/ensemble aggregation	Inform planners on coastal site suitability and model-specific performance for wind energy forecasting. Real-world deployment demands uncertainty handling and adaptive models for climate variability.
18	(Sahin and Ozbay Karakus, 2024)	SVR, ELM, RF Regression, ANN, Weighted k-Nearest Neighbor Regression (WKNNR) (with PCA); CNN/RNN-LSTM (with Genetic Grey Wolf Optimizer (GGWO))	Hydropower Production Forecasting	Daily meteorological and power data from Hirfanlı Dam, Turkey (2007–2021); 64 features, 350,400 data points	The proposed CNN/RNN-LSTM model outperforms all ML models; $r = 0.9802$, $MAE = 0.314$	A hybrid DL model with novel GGWO feature selection achieves state-of-the-art accuracy; a large and rich dataset is used	No uncertainty quantification; lacks real-time adaptive testing; limited to a single dam/location.	Enhances planning for reservoir operation and hydroelectric scheduling, a framework applicable to other renewable domains. The deployment depends on high-quality, frequent data. Model transferability to other regions may be limited.
19	(Altice et al., 2024)	Base DL models: Xception, ResNet-50, VGG-19, AlexNet, custom CNN; Transfer Learning with unfrozen pretrained layers	Visual-based anomaly detection for WT blades	CAI-SWTB (6000 RGB images of indoor & drone-based outdoor shots of small WT blades); external commercial WT blade dataset (516 RGB images)	TL-Xception: 99.92 % on CAI-SWTB, 100 % on large-scale dataset; TL-VGG-19: 99.83 %, custom CNN: 96.67 %	Novel CAI-SWTB dataset; Transfer learning boosts accuracy; generalizes across small & large turbines	Binary classification only; misclassifications due to lighting/glare; limited blur robustness	Enables safe, cost-effective UAV-based inspections and can generalize to other turbine types. Drone-based imaging in varying conditions; real-time edge deployment not tested
20	(Tang et al., 2024)	2D-CNN + LSTM + 3D Gaussian-Frandsen wake model + Rotor-Equivalent Wind Speed (REWS)	Hybrid (ML + Physical modeling)	SCADA data from a real wind farm; Vestas layout for power validation	Achieved low prediction errors; close match to actual power outputs	Integrates wake effect modeling and temporal forecasting. More accurate power estimation using Extreme Gradient Boosting Regression (XGBR)	Does not consider pitch angle control; CNN-LSTM may still be computationally intensive	Supports precise wind farm power forecasting by accounting for layout and wake interactions. Real-time applications may face scalability and latency issues due to 3D modeling and the hybrid approach
22	(Behara and Saha, 2024)	Nature-Inspired Optimization + DL (LSTM, GRU)	Hybrid DL with	CEWRI, NARC Islamabad	RMSE: 3.72 (LSTM), 3.85 (GRU) —	Novel IMF selection (OAIMF); significant accuracy	High computational complexity; exhaustive search	Advances in preprocessing in wind speed

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Table 6 (continued)

S. No	Source	Methods	Category	Dataset / Domain	Results	Advantages	Limitations	Implications and Practical Implementation Challenges
		with Intrinsic Mode Function (IMF) Preprocessing (OAIMF)	Metaheuristic Preprocessing	(2016–2020 atmospheric data)	outperforming wavelet and direct methods	gain over direct & wavelet-based methods; scalable pre-processing; correlation-based feature extraction	(3.77×10^{18}) combinations; limited to a single location case study	forecasting; demonstrates metaheuristic synergy with DL; suitable for high-dimensional, non-linear climate data. Complexity in real-time deployment due to a combinatorial explosion; requires a large training time; parameter tuning overhead
23	(Özdemir, 2024)	Incremental Tuned GAN with 2-level fused discriminator and self-attention	Wind Attribute Analysis, GAN, Unsupervised Learning	Wind energy data (8–12 m/s wind speeds), simulation using MATLAB 2022b	Outperformed ANN, FOPI, fuzzy, CNN, and vanilla GAN in frequency stability, power output, voltage regulation, and torque control	Accurate prediction of wind parameters, improved droop control, robust against fluctuations, adaptive training over time	Computationally expensive, complex model architecture lacks real-world validation	Enhances WT efficiency and grid stability and improves control of DFIG and SG generators. Requires continuous monitoring, large datasets, adaptive control strategy integration, robustness in harsh climates, and ethical AI considerations
24	(Zhao and Zhou, 2024)	Neural Network-Based Ensemble System for Wind Speed Forecasting	Wind Speed Forecasting, ANN, Ensemble Learning	Meteorological data (temperature, humidity, pressure, wind speed) from 5 Eastern Mediterranean sites in Turkey, 1998–2022	MAE: 0.1269, MAPE: 3.074 %, RMSE: 0.1596. The ensemble model significantly outperformed standalone ANN models	High accuracy across multiple sites, scalable with data size, uses real meteorological data, parallel ANN design avoids overfitting and local minima.	High model complexity requires significant training and optimization per site.	Offers scalable and accurate forecasting for real-world wind farms, aiding smart grid integration and planning. Data preprocessing, site-specific network configuration, parallel training, and integration, long training time for large datasets
26	(Mirjalili et al., 2023)	Empirical Mode Decomposition (EMD)-EncoderForest-TCN	Hybrid DL	Wind farm data (China, 2021–2022); additional tests on Northwest and Northeast China wind farms	2.57 % accuracy improvement; up to 5.48 % R^2 increase; Normalized Mean Absolute Error (NMAE)/Normalized Root Mean Square Error (NRMSE) reductions up to 2.65 %	EMD handles nonlinearity and non-stationarity; EncoderForest is robust to noise; TCN models long-term dependencies; modular and adaptable design	Higher computational cost; limited interpretability; sensitive to decomposition and model parameters	Accurate ultra-short-term prediction for grid reliability; adaptable to different wind farms and conditions. Requires > 1 year of clean local data; needs re-training for different sites; Bayesian optimization is computationally expensive

MSE of 0.0023.

Despite these advancements, real-time forecasting under turbulent wind conditions and data drift remains a challenge. High-performing models often demand significant computational resources, while compressed or pruned models may sacrifice accuracy. Furthermore, most evaluations rely on static or site-specific datasets with limited multi-seasonal generalizability. Explainability also remains limited, as many of the most accurate models function as black-box predictors without interpretable outputs.

4.6.3. Fault detection and classification tasks

Fault detection and classification in wind energy systems are vital for

ensuring operational safety, minimizing downtime, and enabling predictive maintenance. These tasks typically involve the use of ML and DL models trained on SCADA data, vibration signals, or acoustic sensor streams. Evaluation metrics such as accuracy, precision, recall, F1-score, and AUC are critical for determining the effectiveness of these models, especially under class-imbalanced conditions where fault cases are rare but critical.

Recent studies present increasingly comprehensive diagnostic metrics, indicating a maturation toward field-deployable solutions. For instance, a CNN–SVM hybrid model in (Yeh et al., 2019) achieved 95 % accuracy on long-cycle maintenance data, while in (Liu et al., 2020), integrated EWT with LSTM, DBN, ESN, and Q-learning, achieving 92 %

accuracy. Similarly, the NARX-QDA-ensemble classifier validated by statistical testing in (Jiménez et al., 2020) surpassed 99 % accuracy.

Studies also highlight the benefits of architectural enhancements through attention mechanisms, imputation strategies, and metaheuristic optimization. For example, (Li et al., 2024) integrated a deep learning-reinforcement learning (DL-RL) model with predictive control, achieving optimization accuracy exceeding 98 %. In another case, (Liu et al., 2023) proposed CorrDQN-FS, which attained the highest prediction accuracy using only four selected features. Hybrid forecasting systems, such as those in (Pradeep et al., 2024) and the CNN-LSTM-Attention architecture in (Jiang et al., 2022), have demonstrated strong capabilities in capturing both local and global patterns in multivariate SCADA data streams.

Lightweight yet effective deployments were also evaluated in (Buestán-Andrade et al., 2024), where CNN, GRU, and Transformer models demonstrated the feasibility of accurate wind forecasting on embedded platforms such as Raspberry Pi. In the context of image-based fault analysis, a hybrid DL model with a novel GGWO-based feature selection method achieved state-of-the-art accuracy using a large and diverse dataset (Sahin and Ozbay Karakus, 2024). The TL-Xception model reported 99.92 % accuracy on the CAI-SWTB dataset and 100 % accuracy on a large-scale industrial dataset (Altice et al., 2024). Similarly, a neural network-based ensemble system in (Zhao and Zhou, 2024) achieved high diagnostic accuracy across multiple sites and showed scalability with increasing data volumes. The EMD-EncoderForest-TCN model in (Mirjalili et al., 2023) reported a 2.57 % improvement in classification accuracy compared to baseline methods.

Despite these advances, several technical challenges hinder widespread deployment. First, cross-site and cross-turbine generalization remains limited, models trained on a specific turbine type or location often degrade in accuracy when transferred to different setups due to variations in SCADA sampling rates, sensor quality, or operational baselines. Second, class imbalance is often inadequately addressed, with many models relying on accuracy as the sole metric, which can mask poor minority-class detection performance. Third, explainability and model transparency remain underdeveloped. Most high-performing models, especially deep or hybrid architectures, act as black boxes, which reduces trust among operators and complicates certification in regulated energy sectors. Fourth, real-time inference and hardware compatibility are often untested in field conditions. Finally, fault-type granularity and taxonomy coverage remain narrow. Most studies focus on a few generic classes and do not explore subtle degradation modes, compound faults, or early pre-fault conditions detectable via latent signals. Richer fault taxonomies, multi-modal sensing fusion, and hierarchical classifiers may be needed for industrial readiness. While current ML/DL-based classifiers achieve high performance under lab conditions, industrial deployment requires models that are not only accurate but also generalizable, explainable, lightweight, and robust to domain shifts.

4.6.4. Control and optimization tasks

Control and optimization tasks in wind energy systems encompass a broad range of objectives, including yaw control, load balancing, battery scheduling, wake steering, and turbine power maximization. These tasks require intelligent decision-making under real-time constraints and often involve highly dynamic, nonlinear, and uncertain operating environments. In recent years, RL and metaheuristic optimization algorithms have emerged as powerful tools to tackle these control challenges, owing to their adaptive, feedback-driven learning mechanisms and ability to handle high-dimensional search spaces.

One of the most prominent trends in this domain is the integration of DRL with control policy learning. Bui et al. (Bui et al., 2020) proposed a multi-agent DRL-based wind farm controller, which improved total energy output by 1.99–4.11 % compared to MPPT strategies. This approach enabled individual turbines to adaptively learn wake interactions and optimize output in partial-load regions through

collaborative policy adjustment. Similarly, Futakuchi et al. (Futakuchi et al., 2021) applied a DQN architecture to wind-battery systems for storage optimization. The model reduced battery dependency and improved storage scheduling by up to 7.75 h, demonstrating the ability of DRL to capture long-horizon dependencies in energy dispatch problems. In the area of yaw control, Liew et al. (Liew et al., 2024) designed a model-free RL controller using RLS that dynamically adjusted turbine yaw angles. Their system reported 11.6 ± 3.0 % gains in power production under partial-wake scenarios without requiring precise wake modelling, highlighting the potential of adaptive policy learning in real-time wind farm operations.

Metaheuristic optimization methods also contribute significantly to control strategies. While RL-based and optimization-aided methods offer transformative potential, several deployment challenges remain. First, the computational cost of training deep policy networks is significant, especially in multi-agent systems or when high-fidelity simulators are unavailable. Second, these models are often sensitive to non-stationary environments, a critical issue in wind energy due to seasonal changes and weather uncertainty. Control and optimization in wind energy systems are increasingly driven by adaptive, learning-based strategies. RL and nature-inspired optimization algorithms offer a path toward autonomous, data-driven control, reducing dependency on analytical modeling.

4.6.5. Critical comparison across techniques

While a broad array of machine intelligence techniques has been applied to wind energy systems, the true value of these approaches lies in their comparative performance under real-world operational constraints. The goal is to assess each technique's suitability for core application areas like forecasting, fault detection, optimization, and control, while accounting for trade-offs in accuracy, interpretability, computational efficiency, and scalability.

Shallow ML models such as SVM, RF, and DT offer interpretability and fast convergence, making them suitable for scenarios requiring lightweight, explainable solutions. These models are well-suited for on-site deployment where hardware resources are limited and real-time responsiveness is essential. However, their limited capacity to capture nonlinear patterns and sequential dependencies reduces their effectiveness in tasks such as long-term forecasting and turbulent wind scenarios.

DL architectures, including CNN, LSTM, GRU, and transformer models, excel in capturing complex temporal and spatial features. These models are particularly effective in forecasting applications where sequential patterns or SCADA data correlations are significant. For example, CNNs are ideal for real-time inference on edge devices due to their parallelizable structure, while LSTM and transformer models are better suited for long-horizon predictions and fluctuating wind profiles. However, deep models typically require extensive datasets, significant computational resources, and often lack interpretability, making them less desirable in regulatory or mission-critical settings unless combined with explainability frameworks.

Ensemble learning approaches, such as gradient boosting and model stacking, provide robust performance under data imbalance and feature variability. These models offer generalization across multiple WT configurations and are less sensitive to noise or overfitting. Their modularity makes them effective in fault detection and multi-sensor fusion tasks. However, ensemble models may not scale efficiently in low-latency environments due to increased inference complexity and higher memory demands.

Hybrid architectures, combining DL with optimization algorithms or signal preprocessing, offer superior adaptability and generalization across diverse conditions. These include models such as CNN-GRU integrated with HHO or VMD-SSA-LSTM-ELM. Such frameworks are highly effective under turbulent wind conditions, non-stationary data, and in scenarios requiring fault-resilient learning. Nevertheless, the integration of multiple subsystems introduces design complexity,

requires longer training time, and may pose challenges for real-time adaptability unless carefully optimized.

Signal decomposition-based methods, including EMD, EEMD, EWT, and SSA, are particularly useful when raw SCADA data is noisy or contains hidden patterns. When combined with LSTM, ELM, or DNN models, these approaches enable more focused learning on isolated frequency or temporal components. They are well-suited for short-term wind ramp forecasting or vibration-based fault detection, but require domain expertise for selecting appropriate decomposition parameters and thresholds to avoid overfitting or information loss.

RL, especially in model-free or deep RL form, is well-positioned for dynamic control and scheduling tasks. These include turbine yaw control, wake steering, and battery energy storage optimization. RL models adapt to changing environmental conditions and can autonomously learn optimal control policies. However, they require large-scale training data, simulation infrastructure, and convergence monitoring, making them challenging for deployment in real-time or safety-critical applications without fallback mechanisms.

Overall, shallow ML models are best suited for rapid diagnostics and interpretability-focused use cases. DL models offer high predictive accuracy for complex forecasting but demand computational and data resources. Ensemble methods are recommended for robust classification under uncertain or imbalanced conditions. Hybrid DL + optimization frameworks are ideal for environments requiring resilience under noise or drift, but require careful integration. Signal decomposition techniques enhance performance where SCADA data is noisy or multiscale in nature. RL is most effective in adaptive control tasks but is less practical for constrained or real-time applications. Model selection should be guided not only by accuracy but by practical deployment criteria such as latency, interpretability, energy efficiency, and generalizability to unseen conditions.

5. Discussion

The study on wind energy prediction using machine intelligence methods has revealed significant findings. This section discusses these findings by classifying them according to the discussed RQs.

5.1. Discussion based on RQ1: What are the current state-of-the-art machine intelligence approaches for enhancing wind energy utilization?

The examined studies applied a wide array of methods, including clustering, DL, and optimization algorithms, to improve tasks such as wind speed prediction, energy price modelling, and condition monitoring (Luo et al., 2018; Crespo-Vazquez et al., 2018; Khan et al., 2019; Khosravi et al., 2018; Santhosh et al., 2019; Zhu et al., 2023; Dhakal et al., 2022). These approaches effectively addressed challenges related to large-scale dataset management and market uncertainty.

ANNs have been widely used for wind power forecasting and multivariate prediction, with several studies demonstrating their role in enhancing forecasting precision and operational efficiency (Hong and Rioflorido, 2019; Mishra et al., 2020; Bui et al., 2020). ML-based regression models have also supported novel applications such as tether force estimation and vulnerability assessment in renewable energy systems (Khan et al., 2019; Liu et al., 2020; Rushdi et al., 2020).

More advanced DL frameworks have emerged for short-term forecasting and ramp event prediction. For instance, Saini et al. (Saini et al., 2020) integrated RNNs with ELM, NARX, and LSTM architectures for wind speed forecasting. Dorado-Moreno et al. (Dorado-Moreno et al., 2020) introduced a multi-task learning framework based on DNN to predict WPRES across multiple wind farms. Liu et al. (Liu et al., 2018) combined EWT with deep networks and RL to create a hybrid forecasting model.

Additional notable efforts include blade delamination detection using NARX and AR models (Jiménez et al., 2020), reactive power forecasting via CNNs (Samet et al., 2020), and district-level electrical

load prediction using DL (Shirzadi et al., 2021). Dhiman and Deb (Dhiman and Deb, 2021) introduced a hybrid method combining discrete wavelet transforms with learning algorithms for ramp event forecasting. Futakuchi et al. (Futakuchi et al., 2021) proposed a DRL-based approach for wind farm operation scheduling, while Hossain et al. (Hossain et al., 2021) developed a hybrid DL model for accurate short-term wind forecasting.

Moreover, techniques such as the Adaptive Differential PSO-Guided Whale Optimization Algorithm (AD-PSO-WOA (Ibrahim et al., 2021)) and RT2FNN have been utilized to enhance control strategies and improve forecasting reliability (Al-Quraan et al., 2023; Tavoosi et al., 2022; Jeong and Kim, 2021).

State-of-the-art ML techniques continue to evolve, with hybrid and DL models demonstrating superior adaptability, precision, and scalability. These methods collectively strengthen wind forecasting systems, reduce uncertainty, and improve the reliability and economic viability of renewable energy integration. Fig. 7 shows the classification of machine intelligence approaches for wind energy enhancement by major key themes based on the reviewed papers.

5.2. Discussion based on RQ2: How can ML techniques be used to optimize WT performance and increase energy production?

Recent studies have demonstrated the effective application of ML techniques to optimize WT performance and maximize energy production. These techniques have enabled improvements in condition monitoring, predictive maintenance, and fault detection, especially when combined with SCADA systems.

Dorado-Moreno et al. (Dorado-Moreno et al., 2020) applied predictive techniques for automated blade inspection using DL, while Diaz et al. (Díaz et al., 2018) utilized MCP methods with ML to estimate long-term WT power output. Sun and Sun (Sun and Sun, 2018) proposed a hybrid RNN framework leveraging SCADA variables for health monitoring, and Yeh et al. (Yeh et al., 2019) employed DL models for forecasting power output in large wind farms.

A range of studies focused on drivetrain and blade fault detection. Bach-Andersen et al. (Bach-Andersen et al., 2018) introduced a DL-based approach using CNNs for drivetrain condition monitoring. Chen et al. (Chen et al., 2018) and Kreutz et al. (Kreutz et al., 2019) addressed blade icing detection by integrating wind and temperature data within SCADA systems using DNN. Shihavuddin et al. (Shihavuddin et al., 2019) enhanced inspection capabilities through drone-assisted image-based DL analysis.

Choi et al. (Choi et al., 2018) used ML models to estimate wind direction from LiDAR-enhanced SCADA data, supporting more accurate yaw control. Saenz (Sáenz Díaz, 2022) applied RL for adaptive yaw angle optimization. Similarly, Liu et al. (Liu et al., 2020) used deep autoencoders to detect blade icing, demonstrating success in identifying multilevel fault features.

Beyond hardware-level diagnostics, several studies proposed ML-driven enhancements in predictive modeling. Kulkarni et al. (Kulkarni et al., 2019) used a combination of LSTM, nonlinear autoregressive networks, and Levenberg–Marquardt algorithms for blade section forecasting. Zhao et al. (Zhao et al., 2018) introduced an early warning system using deep autoencoders, and Qu et al. (Qu et al., 2018) applied GANs for data imputation to address missing or corrupted sensor values.

Control optimization and output forecasting have also been improved. DL-based methods were developed for direct-drive WT fault detection (Teng et al., 2018), power curve modeling with GP (Manobel et al., 2018), and turbine wake prediction using ML-based CFD (Ti et al., 2020; Dehghan Manshadi et al., 2022). These studies consistently demonstrate the scalability of ML frameworks for both local fault diagnosis and system-wide performance forecasting.

ML techniques offer a comprehensive toolset for real-time WT optimization, ranging from fault localization and data reconstruction to intelligent control and long-term forecasting. Their integration with

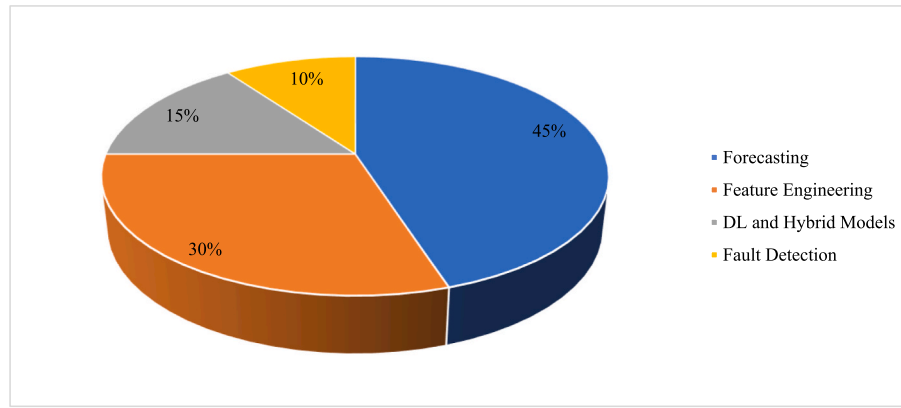


Fig. 7. Classification of machine intelligence approaches for wind energy enhancement by major key themes.

SCADA systems and sensor networks supports improved reliability, reduced downtime, and enhanced energy production.

5.3. Discussion based on RQ3: What are the main challenges and limitations associated with the integration of machine intelligence approaches in wind energy systems?

The integration of ML into wind energy systems holds considerable potential for improving efficiency, fault detection, and predictive maintenance. However, this process is not without significant technical and operational challenges. A primary barrier lies in the compatibility of ML models with legacy infrastructure. Many existing wind energy systems were not originally designed to support data-intensive, adaptive algorithms, and retrofitting such systems often demands costly upgrades to both hardware and control software.

Another persistent limitation is the requirement for high-quality, context-specific data. Effective ML models depend on accurate, voluminous datasets for training. Yet, WTs often generate heterogeneous and noisy data streams that require significant preprocessing. Incomplete SCADA logs, sensor faults, and environmental disturbances further complicate data quality (Jalali et al., 2021; Deepa and Banerjee, 2022; Yang et al., 2022; Chen et al., 2021). These factors restrict model accuracy and increase the likelihood of overfitting or false alarms.

The complexity and nonlinearity of wind energy systems also present challenges. As highlighted in previous studies (Liu et al., 2020; Sharifzadeh et al., 2019; Yang et al., 2022; Das et al., 2024), turbines operate under constantly fluctuating atmospheric conditions—including wind speed, turbulence, humidity, and temperature. ML models must not only capture these temporal dynamics but also generalize across turbine types, locations, and operational contexts. This complexity hinders model transferability and limits real-world scalability.

Moreover, interpretability and trust in ML-driven decision-making remain critical issues. Operators often require transparent, explainable models, yet many high-performing approaches, such as DL, function as black boxes. Without clear interpretability, system operators may hesitate to adopt or act on model recommendations.

Lastly, the resource-intensive nature of ML model development and deployment should not be underestimated. Developing robust models involves considerable computational overhead, specialized expertise, and continuous maintenance. For smaller operators, the upfront costs and lack of in-house technical capacity can be prohibitive.

While ML integration into wind energy systems offers measurable benefits, it also introduces a layered set of challenges. These include infrastructure compatibility, data quality, system complexity, model explainability, and cost of implementation. Addressing these limitations is essential to scale intelligent wind energy solutions from pilot studies to industry-wide adoption.

5.4. Discussion based on RQ4: What are the potential benefits and drawbacks of using machine intelligence approaches in wind energy utilization compared to traditional approaches?

This section evaluates the comparative strengths and limitations of machine intelligence approaches—including ML, DL, RL, and hybrid models—against traditional statistical and physical models in wind energy applications. Various algorithms such as SVM, KNN, RF, DT, regression models, LSTM, and CNN have been widely employed. Their hybridization with boosting and nature-inspired algorithms has further expanded their efficacy across forecasting, optimization, and control.

Wind energy predictions are influenced by environmental factors like temperature, wind direction, data characteristics, for example, quality and historical depth, and methodological parameters like feature selection, training window. The literature discussed emphasizes the need to account for seasonal wind variability and historical temperature correlation when optimizing training and validation frameworks (Pandit and Infield, 2019; Sun and Sun, 2018; Lv and Wang, 2023; Singh and Rizwan, 2022).

DL algorithms such as LSTM and CNN demonstrate strong capabilities in spatiotemporal correlation and feature generalization, while ANNs have been effective in producing high-resolution forecasts. Studies also show that ensemble learning improves prediction robustness by integrating multiple learners. For instance, hybrid combinations of machine intelligence algorithms are the most prevalent (40 %), followed by hybrids involving parameter tuning (32 %) and those integrating machine vision and boosting strategies (22 %) (Fig. 8).

A hybrid combination of machine intelligence methods dominates wind energy optimization tasks, especially in forecasting and fault prediction. RL, particularly when combined with ANN or DL, has been used for real-time job scheduling, control optimization, and yaw alignment. Transfer learning and adaptive parameterization further enhance the adaptability of DL models to evolving turbine conditions. ELMs with robust outlier handling improve short-term wind speed forecasting by tuning hidden layer parameters. Moreover, ANN, SVM, and GP models have successfully been deployed to model energy output and grid-side uncertainties.

The use of advanced algorithms such as DNN and Density-based Grey Artificial Bee Colony algorithm (D-GABC) has enhanced fault prediction and system diagnostics. LSTM can handle irregular time-series patterns, enabling more accurate wind speed forecasting. Collectively, these strategies contribute to efficient energy conversion, improved predictive control, and intelligent diagnostics, pushing wind energy systems closer to real-time responsiveness and smart grid integration.

Despite these advantages, the application of ML in wind energy still faces notable limitations:

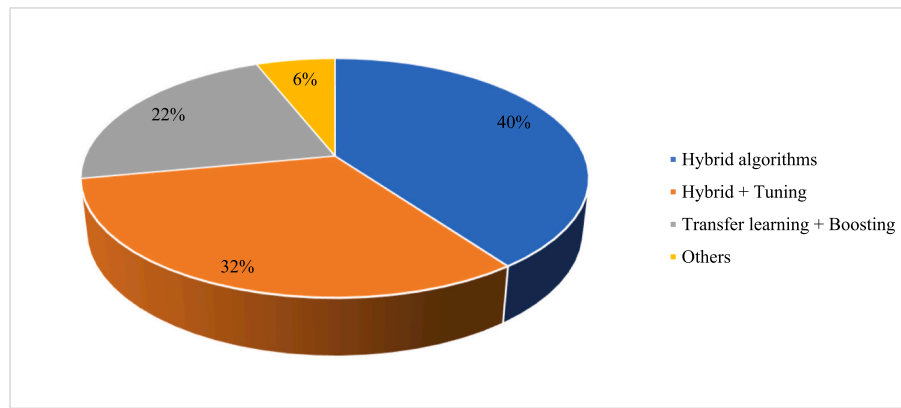


Fig. 8. Distribution of machine intelligence methods applied in wind energy.

- SVM, KNN, and regression models often require large and well-labeled datasets. Model accuracy may degrade in the presence of noise or imbalance.
- CNNs and DL architectures require substantial computational resources, limiting their feasibility for deployment in small-scale or resource-constrained environments.
- While effective, many DL models operate as black boxes, complicating efforts to explain outputs or transfer them across turbine types and geographic conditions.
- Over-reliance on specific datasets can reduce generalizability. Hybrid models with boosting or nature-inspired components require careful tuning of hyperparameters.
- The lack of standardized methods in SCADA-based monitoring hampers cross-study comparisons. Real-world validation remains limited in many ML-driven solutions.
- ML algorithms can inherit biases from historical training data and may underperform under unseen environmental conditions.

The applicability of these machine intelligence techniques varies across wind energy subdomains. As shown in Fig. 9, DL and hybrid models are predominantly applied in wind speed and wind power forecasting, while RL is more common in energy management and control tasks. Other methods are also sporadically applied in domains like wind farm design, predictive maintenance, and fault diagnosis. DL and hybrid techniques are heavily utilized in forecasting tasks, while RL and other methods support specialized domains such as energy

management and fault diagnosis.

Although substantial progress has been made, the deployment of machine intelligence techniques in the wind energy sector remains uneven. The majority of the studies reviewed were validated under real-world conditions, including SCADA records, operational wind farms, offshore wind conditions, and atmospheric measurements. Nevertheless, many approaches continue to rely heavily on synthetic data, simulated environments, or restricted historical datasets, limiting their robustness under dynamic field conditions. Key challenges identified include high computational requirements, sensitivity to site-specific conditions, model overfitting, limited generalization across geographical regions, and the absence of robust uncertainty quantification mechanisms. Furthermore, many promising DL and RL models still struggle with practical scalability and latency issues in real-time operational contexts. Therefore, future research should prioritize lightweight, energy-efficient, and interpretable models, stronger real-time validation frameworks, adaptive online learning strategies, and the integration of physics-informed or hybrid systems that can handle the inherent uncertainty and variability of renewable energy environments to achieve broader and more reliable adoption in optimizing wind farm operations. Table 8 critically evaluates machine intelligence models for wind energy by categorizing them into hybrid, ML/DL, RL, or heuristic-based techniques. It systematically identifies real-world deployment outcomes, highlights operational gaps, and addresses the need for synthesized insights across diverse modeling approaches to overcome current fragmentation in the field.

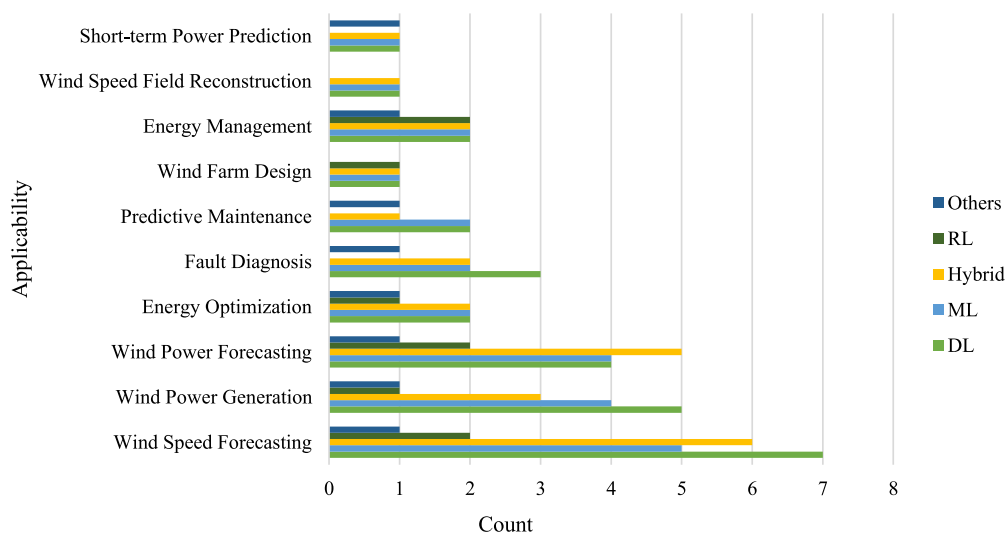


Fig. 9. Applicability of machine intelligence methods across wind energy domains.

Table 8

Assessment of machine intelligence and hybrid models in wind energy: real-world deployment and research challenges.

S. No	Model/Technique	Model Type	Real-world Deployment	Research Gaps	Source
1	NARX + AR Models for Blade Monitoring	ML/DL (traditional time series models)	Excel (Real WT blade tests)	Limited to delamination; no generalization to other faults	(Jiménez et al., 2020)
2	DRL for Real-Time Control	RL	Partially Excel (Synthetic signals tested)	Real atmospheric conditions vary; they lack long-term stability tests	(Vijayshankar et al., 2021)
3	Hybrid PCA + DL for Forecasting	Hybrid (Feature extraction + DL)	Excel (Historical datasets)	Sensitivity to feature extraction parameters	(Khan et al., 2019)
4	Pinball Loss Optimization + SVR	ML/DL (Classical SVR with loss optimization)	Excel (Better pinball loss scores)	Only a short-term horizon validated	(Sun et al., 2019)
5	Hybrid DL with RL Integration	Hybrid (DL + RL integration)	Excel (Improved wind speed accuracy)	Only tested on fixed datasets	(Liu et al., 2020)
6	CNN for Reactive Power	ML/DL (CNN)	Excel (SCADA from the working wind farm)	Limited to RC	(Samet et al., 2020)
7	MLP + SVR + CatBoost	Hybrid (Multiple classical ML models ensemble) integrated with Renewable Energy Sources (RES)	Excel (Real operational data)	No feature importance ranking	(Khan et al., 2020)
8	DL for District Load Forecast	ML/DL (DL prediction)	Excel (Utility-grade load datasets)	Rural/urban bias	(Shirzadi et al., 2021)
9	DRL-Based DeepComp Error-Compensable Forecasting	RL (Deep Reinforcement Learning)	Excel (Wind + Solar)	Limited to small-scale testbeds	(Jeong and Kim, 2021)
10	Hybrid CNN-LSTM with Whale Optimization	Hybrid (DL + heuristic optimization)	Partially Excel (Coastal and offshore testing)	Computationally expensive for online use	(Nezhad et al., 2021)
11	Ensemble Averaging Wind Forecast	ML/DL (Classical ensemble learning)	Excel (Operational farm datasets)	Only short-term predictions	(Piotrowski et al., 2022)
12	RF + LSTM-RNN	Hybrid (Ensemble of RF + DL)	Excel (Meteorological station data)	No long-term trend analysis	(Salah et al., 2022)
13	DCNN for Spatiotemporal Wind Speed Forecasting	ML/DL (DCNN)	Partially Excel (Multi-site data)	Limited generalization across climates	(Zhu et al., 2018)
14	Hybrid ANFIS-PSO for Wind Speed Prediction	Hybrid (Fuzzy inference + heuristic optimization)	Excel (Higher prediction accuracy)	Model complexity limits real-time adaptability	(Khosravi et al., 2018)
15	Deep Autoencoder for Blade Icing Detection	ML/DL (Autoencoder – Unsupervised DL)	Excel (SCADA data validated)	SCADA data noise affects model robustness	(Liu et al., 2018)
16	Hybrid DL for Smart Grids	Hybrid (Multi-branch DL structure)	Excel (Smart grid field data)	Edge device resource limitations	(Santhosh et al., 2019)
17	DWT + FFT + DCNN	Hybrid (Signal processing + DL)	Partially Excel (Wind farm time series)	Preprocessing overhead	(Mishra et al., 2020)
18	CFD + Stochastic ML Offshore	Hybrid (Physics-based + stochastic ML)	Excel (Offshore farm data)	Computationally intensive	(Richmond et al., 2020)
19	Twin SVR + CNN for Ramp Event Prediction	Hybrid (classical ML + DL fusion)	Excel (Real ramp events)	Limited to onshore; offshore dynamics not considered	(Dhiman and Deb, 2021)
20	CNN + GRU + HHO	Hybrid (DL + heuristic optimization)	Excel (Short-term real wind farm)	Hyperparameter tuning challenges	(Hossain et al., 2021)
21	Multi-Agent DRL for Wake Reduction	RL (Deep Reinforcement Learning)	Partially Excel (Simulation studies)	Stability during cooperative learning	(Bui et al., 2020)
22	Scheduled DRL for Ramp Events	RL (Deep Reinforcement Learning)	Excel (Working farm battery integration)	Limited battery modeling	(Futakuchi et al., 2021)
23	DNN for Wind Speed Extrapolation	ML/DL (Deep Neural Networks)	Partially Excel (LiDAR ground-truth only)	Limited high-altitude generalization	(Mohandes and Rehman, 2018)
24	DRL for Cyber Vulnerability Scoring	RL (Deep Reinforcement Learning)	Partially Excel (Simulated power grids)	Cyber-physical validation missing	(Liu et al., 2020)
25	MCP + ML Approaches	Hybrid (Time series + ML methods)	Partially Excel (Simulated MCP models)	Ignored environmental variables	(Díaz et al., 2018)
26	LSTM for Fluid-Structure	ML/DL (LSTM-based simulation)	Partially Excel (Vortex Bladeless WT Simulations)	Single design validation	(Dehghan Manshadi et al., 2022)
27	AGA-DDPG (Adaptive GA + DDPG)	Hybrid (Evolutionary Optimization + DRL)	Partially Excel (Simulated MEC offshore WT + UAVs)	Idealized assumptions affect real-world adaptation	(Song et al., 2024)
28	DL-RL-MPC (RBFN + DQN + MPC)	Hybrid (DL + RL + Predictive Control)	Partially Excel (Simulated OpenFAST 15-MW turbine)	Heavy computation; a large number of training steps are needed	(Li et al., 2024)
29	DRL with Wind Power Prediction Intervals	Hybrid (DL + RL)	Excel (Real wind farm)	High computational cost; generalizability untested	(Liu et al., 2023)
30	Double DQN (DDQN)-based Non-causal Control with Wave Prediction	Hybrid (DRL + non-causal control)	Excel (Real Cornwall wave data + Synthetic waves)	Prediction quality sensitive to noise & model horizons	(Wang et al., 2024)
31	GAT-GRU-TCN-DDPG	Ensemble DL + RL	Excel (Wind Farms Data)	Requires predefined graph structure; feature tuning	(Chengqing et al., 2023)
32	CorrDQN-FS	DRL for Feature Selection	Excel (Commercial office building energy data)	Hyperparameter tuning overhead	(Liu et al., 2023)
33	Meta-MARL (Meta Multi-Agent RL)	DRL (Multi-agent + Meta-RL)	Excel (IEEE 39-bus with real PV, wind, load data)	Second-stage optimization limitations	(Li et al., 2024)
34	Hybrid Forecasting (K++ + TLBO + Deep SARSA)	Hybrid (Clustering + Optimization + DRL)	Partially Excel (Hybrid Renewable Forecast Simulation)	No detailed quantitative evaluation is available	(Pradeep et al., 2024)

(continued on next page)

Table 8 (continued)

S. No	Model/Technique	Model Type	Real-world Deployment	Research Gaps	Source
35	Baseline to Residual LSTM for Wind Prediction	ML/DL (Comparative Study)	Excel (Real-world historical wind data)	Only MAE used; lacks diverse metric evaluation	(Anushalini and Sri Revathi, 2024)
36	CNN/GRU/Transformer for SCADA Forecasting	ML/DL (Time-series prediction)	Excel (SCADA data)	High GRU energy use; transformer accuracy issues	(Buestán-Andrade et al., 2024)
37	LightGBM with MLOps for SCADA Forecasting	ML (Tree-based with MLOps)	Excel (SCADA Data with MLOps pipeline)	Domain-specific feature engineering is needed	(Oyucu and Aksöz, 2024)
38	RF Pseudo-2D RANS Model & DeepHyper Optimization	Hybrid (Physics + ML optimization)	Excel (Oklahoma Wind Farm SCADA Data)	Wake interaction under/overestimation challenges	(Moss et al., 2024)
39	DT/Ensemble Methods for Wind Speed	ML (DT, Ensembles)	Excel (Coastal Wind Sites Historical Data)	No uncertainty quantification; regional limitation	(Mollick et al., 2024)
40	Hybrid CNN-RNN-LSTM with GGWO	Hybrid (DL with Evolutionary feature selection)	Excel (Hirfanlı Dam Hydropower Data)	Single location tested; real-time adaptation lacking	(Sahin and Ozbay Karakus, 2024)
41	Transfer Learning for Blade Anomaly Detection	ML/DL (Transfer Learning)	Excel (CAI-SWTB, Outdoor Drone Blade Images)	Limited to binary classification; glare/blur issues	(Altice et al., 2024)
42	2D-CNN + LSTM + Wake + REWS Model	Hybrid (Physics-based + DL)	Excel (Real Wind Farm SCADA + Vesta's layout)	No pitch control modelled; computational load concerns	(Tang et al., 2024)
43	Nature-Inspired Optimization + DL (OAIMF)	Hybrid (Metaheuristic + DL)	Partially Excel (Islamabad CEWRI data)	Extremely high computation for the IMF search; single site	(Behara and Saha, 2024)

5.4.1. Barriers to real-world deployment: SCADA Integration, regulatory gaps, and sustainability trade-offs

Despite substantial advancements in machine intelligence techniques for wind energy applications—ranging from predictive maintenance and power forecasting to smart control—real-world deployment remains limited and fragmented. Many models reviewed in this study report validation using SCADA data or operational wind farm environments; however, deployment at scale still faces critical obstacles. These stem not only from algorithmic complexity but also from data infrastructure, regulatory misalignment, cybersecurity vulnerabilities, economic constraints, and sustainability trade-offs. Bridging these gaps is essential to transition from proof-of-concept models to robust, field-deployable systems.

SCADA system integration: from raw data to deployable insights

A primary barrier to implementation lies in the integration of ML models with SCADA systems. These systems, while widely deployed across wind farms, often operate with heterogeneous sensor protocols, non-standard formats, and low-resolution data. SCADA data is frequently noisy, incomplete, or temporally misaligned due to sampling delays, resulting in data integrity issues that compromise ML model reliability.

For instance, Maldonado-Correa et al. (Maldonado-Correa et al., 2024) employed transformer networks to detect Insulated Gate Bipolar Transistor (IGBT) faults months in advance, but only after conducting extensive preprocessing—including feature selection and time-series synchronization—to address SCADA irregularities. Likewise, Wang et al. (Wang et al., 2025) improved WT power estimation by filtering SCADA input using pitch angle thresholds and rotor speed constraints, eliminating aberrant readings due to turbulence or communication lag.

These scenarios emphasize the need for standardized preprocessing pipelines tailored to SCADA's operational context. Key strategies for improving the reliability of ML models in wind energy systems involve several data preprocessing techniques. Noise filtering is commonly performed using statistical thresholds or wavelet transforms to eliminate erratic sensor readings. Missing data are addressed through imputation methods that consider temporal dependencies, such as GP models, which offer time-aware estimations. To mitigate the impact of high-dimensional, redundant SCADA inputs, dimensionality reduction techniques like PCA or autoencoders are employed. Additionally, temporal alignment through resampling or interpolation ensures that asynchronous sensor data are synchronized with control loop cycles, thereby improving model input consistency.

Additionally, the development of “SCADA-aware” ML toolkits—with plug-and-play compatibility across turbine vendors—can streamline integration. These should encapsulate domain-specific heuristics like turbine cut-in speeds, wake models, and interface directly with edge

computing devices for real-time inference.

Regulatory and cybersecurity constraints

Regulatory compliance poses a second major barrier. ML models used in power systems must satisfy stringent grid codes, which dictate real-time response limits, power ramping constraints, frequency regulation protocols, and blackout recovery standards. However, most ML models are designed for predictive accuracy, not for certifiability or traceability, which hampers deployment in grid-critical applications.

For instance, Liu et al. (Liu et al., 2020) proposed a DRL-based cybersecurity scoring system using a CVSS-aligned transition model to assess wind power system vulnerabilities. Yet, its performance was only evaluated under simulation, not in live grid environments, where real-time grid stability must be preserved during cyber intrusions. Moreover, explainability and auditability—core requirements in safety-critical systems—remain underdeveloped in most AI models for wind energy.

Concrete solutions to enhance the trustworthiness and deployment readiness of ML in wind energy systems involve both technical and regulatory innovations. The integration of Explainable AI frameworks—such as SHapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), and Layer-wise Relevance Propagation—enables transparent decision-making, allowing stakeholders to understand model outputs and validate their reliability. Collaborative efforts with grid operators and regulatory bodies are essential to develop standardized certification protocols for AI applications in power systems, ensuring compliance and operational safety. Furthermore, the use of Digital Twins allows for virtual simulations of grid behavior under extreme or faulty conditions, providing a testbed to validate AI models before real-world deployment. To address cybersecurity and data privacy concerns, techniques like federated learning and adversarial training are employed, enabling robust model development without directly exposing sensitive SCADA data, as demonstrated in recent studies (Skrodelis et al., 2024; Aydin, 2025). These approaches must be formalized within regulatory sandboxes or testbeds, enabling iterative validation before field integration.

Economic scalability and infrastructure requirements

Economic feasibility is a third bottleneck. High-performing models such as Perturbed RF (Irfan et al., 2025) or Huber Adam Regression Optimizer (HARO) (Raju et al., 2025) often require large-scale computation, GPU clusters, and frequent retraining cycles. These demands are manageable for utility-scale wind operators but prohibitively expensive for small- to mid-size wind farms.

For example, Piotrowski et al. (Piotrowski et al., 2022) demonstrated the value of hybrid ensemble models incorporating turbine power curves and NWP features, achieving strong forecasting accuracy. However, their model required managing 81 input variables across 29

months of data, far beyond what most decentralized wind installations can afford to compute or store.

To address cost barriers in deploying machine intelligence within wind energy systems, research should emphasize resource-efficient solutions tailored for embedded and edge environments. Lightweight architectures such as MobileNet, TinyLSTM, and quantized CNNs are well-suited for deployment on low-power devices without compromising performance. On-device training and inference can be facilitated through platforms like TensorFlow Lite and NVIDIA Jetson, enabling real-time analytics without reliance on high-cost infrastructure. Edge computing strategies further reduce operational costs by allowing models to perform inference locally, transmitting only critical insights to centralized servers, thereby minimizing bandwidth and cloud dependency. Additionally, transfer learning can significantly lower training costs by enabling the adaptation of pre-trained models to site-specific conditions using limited local data, thus reducing the need for large-scale data collection and extensive retraining. Furthermore, cost-benefit analysis tools should be developed to evaluate model performance versus hardware cost, helping wind farm operators select the most appropriate intelligence solution for their budget and scale.

Environmental impact of AI lifecycle in wind energy

An increasingly relevant concern is the environmental sustainability of machine intelligence pipelines. Many studies overlook the carbon footprint associated with model training, tuning, and deployment, particularly when models are retrained frequently due to site-specific variability or concept drift.

Da Silva et al. (Da Silva, 2025) recently emphasized that frequent retraining of wind forecasting models in highly variable environments significantly increases energy use and emissions, counteracting the green energy mission. Models that rely on exhaustive hyperparameter searches, ensemble averaging, or long-sequence transformers are especially energy-intensive.

To minimize the environmental cost of deploying machine intelligence in wind energy systems, researchers should adopt sustainability-aware modeling practices. This includes systematically measuring and reporting energy consumption and associated CO₂ emissions during model training, alongside traditional performance metrics such as accuracy (Liu et al., 2025; Shahzad et al., 2025; Ayogu et al., 2025; Mamyrbayev et al., 2025). Leveraging low-carbon cloud infrastructure or energy-efficient hardware, such as Tensor Processing Units, can further reduce the carbon footprint of model development. Sparse modeling techniques—such as pruning, knowledge distillation, and low-rank approximation—enable significant reductions in model size and computational load without sacrificing predictive performance. Moreover, incorporating *meta-learning* and continual learning frameworks allows models to incrementally adapt to new data by updating only select components, thereby preserving computational resources and reducing the need for full retraining.

Green AI metrics should become part of model benchmarking standards, particularly for public-sector and sustainability-focused wind energy projects. To overcome real-world deployment barriers of machine intelligence in wind energy, a multi-pronged strategy is essential. Technically, models must become more robust to noisy SCADA data, more adaptable to deployment-scale constraints, and more transparent in decision-making. Regulatory compliance and cybersecurity resilience must be embedded into system design, not treated as post hoc additions (Liu et al., 2025; Shahzad et al., 2025; Ayogu et al., 2025; Mamyrbayev et al., 2025). Economically, models should be optimized for edge computing, minimal training, and operational cost efficiency. Environmentally, the carbon cost of intelligent models should be monitored and minimized through green AI practices (Liu et al., 2025; Shahzad et al., 2025; Ayogu et al., 2025; Mamyrbayev et al., 2025).

Ultimately, success will depend on interdisciplinary collaboration among ML researchers, turbine manufacturers, grid operators, policy-makers, and environmental engineers. Only through a unified effort can machine intelligence evolve from lab-scale experimentation to scalable,

sustainable, and trustworthy deployment in the wind energy sector.

5.5. Limitations, future directions, and practical applications

This review is limited to studies published in English, which may introduce a language bias and restrict the global comprehensiveness of the findings. Additionally, the paper primarily focuses on machine intelligence approaches for enhancing wind energy systems, excluding several critical dimensions such as policy, economic factors, and infrastructure development. Other important aspects, such as WT design, cybersecurity, and grid integration strategies, fall outside the scope of this review and are only briefly discussed, but they represent valuable directions for future research.

Some of the future directions are as under:

1. There is a need to design experimental setups and simulation frameworks for assessing the stability and performance of intelligent models within integrated electrical networks. This includes incorporating uncertainty quantification, load balancing, and fault-tolerant mechanisms.
2. Developing robust ensemble learning models and leveraging advanced data fusion techniques can significantly enhance the accuracy and reliability of wind energy forecasting. In particular, integrating remote sensing data with multi-source inputs such as SCADA systems, meteorological sensors, and satellite imagery may improve both the spatial and temporal resolution of forecasts.
3. Advanced signal decomposition methods and temporal feature extraction strategies, such as wavelet transforms and recurrent neural architectures, are essential to capture the dynamic nature of wind patterns, thereby improving predictive and control capabilities.
4. Combining ML models with boosting algorithms and nature-inspired optimization techniques can yield adaptive and self-optimizing systems for applications such as predictive maintenance, energy dispatch, and fault detection. These hybrid methods are particularly well-suited for real-time risk assessment, anomaly detection, and system resilience enhancement.
5. Future studies should incorporate explainability techniques to interpret model decisions and foster trust, especially in safety-critical environments.
6. Research should assess model robustness across different geographies, terrains, and climate conditions, developing transfer learning strategies to adapt models to varying environmental settings with minimal retraining.

To apply these advances in real-world settings, the following implementation challenges need to be resolved:

1. It is vital to design lightweight, resource-efficient ML/DL architectures that support real-time inference. Techniques such as model pruning, quantization, and edge AI deployment can reduce computational costs and energy consumption.

Addressing the limitations of training data, especially in data-poor or underrepresented regions, requires the adoption of transfer learning, synthetic data generation, and domain adaptation methods to ensure model generalizability and robustness.

2. Scalable deployment strategies should include containerized, modular architectures compatible with existing SCADA systems. These should feature automated model retraining, continuous monitoring, version control, and seamless cross-platform integration to support long-term, dynamic operation.
3. Establish lifecycle protocols for data acquisition, model training, validation, deployment, monitoring, and retirement to maintain performance and mitigate drift over time.

4. Promote the development of human-in-the-loop systems and provide training programs for operators and engineers to effectively interpret AI recommendations and manage AI-integrated systems.

6. Conclusions

The integration of machine intelligence into wind energy systems is playing a transformative role in optimizing key operational objectives. Advanced data-driven models—including ML, DL, RL, and hybrid frameworks—are significantly enhancing forecasting accuracy, control responsiveness, and fault detection across various components of wind energy infrastructure. At the core of these advancements lies the goal of system optimization: maximizing energy output, improving component reliability, aligning power curves with expected performance, and reducing operational costs. Techniques such as predictive maintenance, adaptive control, and fault-tolerant dispatch demonstrate how intelligent algorithms can outperform conventional rule-based methods, particularly under the constraints of noisy and irregular SCADA data. Hybrid approaches that combine signal decomposition, evolutionary optimization, and DNNs have proven particularly effective in managing uncertainty and supporting real-time, data-driven decision-making. Despite these gains, challenges remain in real-world deployment, including integration with heterogeneous SCADA systems, limited model interpretability, high computational demands, and insufficient validation across diverse grid and environmental conditions.

To address these limitations and scale machine intelligence in wind energy optimization, future research should emphasize lightweight, energy-efficient architectures for edge deployment; interpretable models that meet regulatory requirements; and transferable frameworks adaptable to varying geographies and turbine types. Additionally, continuous learning mechanisms that require minimal retraining can support long-term operational adaptability.

A comprehensive optimization strategy must also account for broader sustainability metrics, including emissions from model training, system resilience, and economic feasibility. Intelligent solutions should be evaluated not only on performance metrics but also on their environmental and economic impacts over the system lifecycle. Ultimately, machine intelligence is reshaping wind energy systems by enabling predictive, autonomous, and adaptive operations. With continued interdisciplinary research and robust validation practices, these technologies hold the potential to unlock scalable, cost-effective, and sustainable solutions for future wind energy development.

CRediT authorship contribution statement

Ashutosh Kumar Dubey: Conceptualization, Methodology, Formal analysis, Investigation, Software, Data curation, Writing – original draft, Visualization, Validation. **Abhishek Kumar:** Conceptualization, Methodology, Formal analysis, Investigation, Software, Data curation, Writing – original draft, Visualization, Validation. **Isaac Segovia Ramírez:** Conceptualization, Methodology, Formal analysis, Investigation, Software, Data curation, Writing – original draft, Visualization, Validation. **Fausto Pedro García Márquez:** Data curation, Writing – original draft, Validation, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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